

SaNaML

Common Statistical Tests: a step-by-step manual with practice data

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SaNaML Statistics & Machine Learning • 0.10.2 Home Data **Analyses** Charts Power analysis Clustering Machine learning Word report

Common statistical analysis

Test selection guide (Help me choose)
Answer two questions and SaNaML suggests the right test.

Descriptive statistics
Means, SDs, medians, skewness and kurtosis, overall or by group.

Independent-samples, paired-samples and one-sample t-tests (Compare two groups or two measurements)
Welch and Student t-tests, with the Mann-Whitney U and Wilcoxon signed-rank tests.

One-way ANOVA (Compare three or more groups)
Welch ANOVA, Tukey and Games-Howell post hoc tests, Kruskal-Wallis test with Dunn.

One-way repeated-measures ANOVA (Same people, three or more times)
Mauchly test with Greenhouse-Geisser and Huynh-Feldt corrections; Friedman test.

Independent-samples, paired-samples and one-sample t-tests (Compare two groups or two measurements)

What do you want to compare?

- ☒ **Independent-samples t-test (Two groups of different people)**
For example women vs men, or users vs non-users, on the same score.
- ☐ **Paired-samples t-test (The same people measured twice)**
For example satisfaction before and after, or two ratings by each respondent.
- ☐ **One-sample t-test (One score against a fixed value)**
For example: is the average agreement above the scale midpoint of 4?

Score to compare
Exam_score
A scale variable: an item, a construct score or another number.

Groups
Gender (2 groups)
A categorical variable. If it has more than two groups, choose the two to compare below (or use One-way ANOVA).

Direction of the test

- ☒ **Two-tailed test (Any difference)**
The usual choice: a difference in either direction counts.
- ☐ **Right-tailed (one-tailed) test (Only "first is greater")**
Use only if your hypothesis predicted this direction before seeing the data.
- ☐ **Left-tailed (one-tailed) test (Only "first is smaller")**
Use only if your hypothesis predicted this direction before seeing the data.

Independent-samples t-test (Comparing Exam_score between female and male)

What it means: female scored higher than male on Exam_score (difference 4.19 points). The difference is statistically significant and its size is small ($|d| = 0.44$).

Worked example: *practice_students.csv*, 400 simulated students

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About this manual

This manual teaches the common statistical tests with SaNaML, the statistics and machine learning program of the SaNaSoft suite. It follows one practice data file, `practice_students.csv`, from opening the file to a finished Word report: data checking, computing and recoding variables, descriptive statistics and charts, t-tests, analysis of variance, repeated measures, chi-square tests, correlation and regression, power analysis, and a short machine learning check on a regression with cut-offs for a yes/no outcome. Every chapter says what to click, what you will see, which rule of thumb applies, how to read the result and how to report it. All screenshots and numbers come from real runs of SaNaML 0.10.2, so you can follow along and obtain the same results.

Who this manual is for

Postgraduate students and researchers who need to choose, run and report common statistical tests. You do not need to know any statistics software. Each statistical idea is explained in plain language in a "Concept" box when it first appears; if you already know the idea, skip the box. The chapters can be read in order, or you can go straight to the test you need using the contents list and the test-choice table in Appendix B.

The practice data are simulated

The file `practice_students.csv` describes 400 students of a made-up university. **The data are simulated.** No real person appears in them, and the effects in the file were planted on purpose (for example, women score a little higher than men, and a flipped classroom raises the post-test). You can therefore check whether your analysis finds what was built in. Do not report these numbers as findings about real students. The exercises at the end of most chapters use the other practice files that ship with SaNaML (see "The practice files" below).

What you need

- A computer with an up-to-date browser (Google Chrome, Microsoft Edge, Mozilla Firefox or Safari). SaNaML runs inside the browser; your data are read on your own computer and are not uploaded.
- The practice file `practice_students.csv` and, for the exercises, `practice_marketing.csv`, `practice_brand_survey.csv`, `practice_hr.csv`, `practice_engagement.csv`, `segments_demo.csv` and `careless_demo.csv`. Your own data should be a CSV, .xlsx spreadsheet or .sav / .zsav file with one row per person and one column per variable, with the column names in the first row. Missing values are blank cells.
- Roughly one afternoon: a chapter takes between 20 and 40 minutes if you repeat each step.

How to read the instructions

Table 1

Conventions used in this manual

Convention	Meaning	Example
Bold green text	A button, tab, menu item, box or list on the screen. Click or choose exactly this.	Run the comparison
→	A sequence of clicks.	Data → Choose a file...
Numbered steps	Actions to carry out in this order.	1. Open ... 2. Click ...
<i>Italics</i>	A technical term explained nearby, or text exactly as it appears in the results.	<i>Welch's t-test</i>

The practice files

Table 2

The practice files that ship with SaNaML

File	What it describes	Used in this manual
practice_students.csv (400 rows)	A university study: tests, ANOVA, regression, reliability, data quality, machine learning	All chapters
practice_marketing.csv (1,500 rows)	A retailer's customers: A/B test, spend, churn	Exercises in chapters 5, 9, 10 and 11
practice_brand_survey.csv (500 rows)	A brand perception survey, 1 to 7 ratings	Exercises in chapters 1, 2 and 6
practice_hr.csv (1,200 rows)	A company's employees: pay, training, attrition	Chapter 6 (Games-Howell) and exercises
practice_engagement.csv (600 rows)	An engagement survey, three scales	Exercises in chapters 1 and 6
careless_demo.csv (300 rows)	A survey with planted careless responders	Exercise in chapter 2
segments_demo.csv (360 rows)	Customer segments	Charts exercise in chapter 3

Each file has an identifier column (StudentID, CustomerID and so on). It is only a label; never use it as a variable.

Feedback on this manual is welcome at sabraz@seu.ac.lk.

1. Getting started

1.1 Open SaNaML

Open SaNaML in your browser at the address of your installation. It opens on its Home screen (Figure 1). The header shows the program name with its version ("Statistics & Machine Learning · 0.10.2") and the main menu: **Home**, **Data**, **Analyses**, **Charts**, **Power analysis**, **Clustering**, **Machine learning** and **Word report**. Once you have added results to the report, the last item shows their number, for example **Word report (3)**.

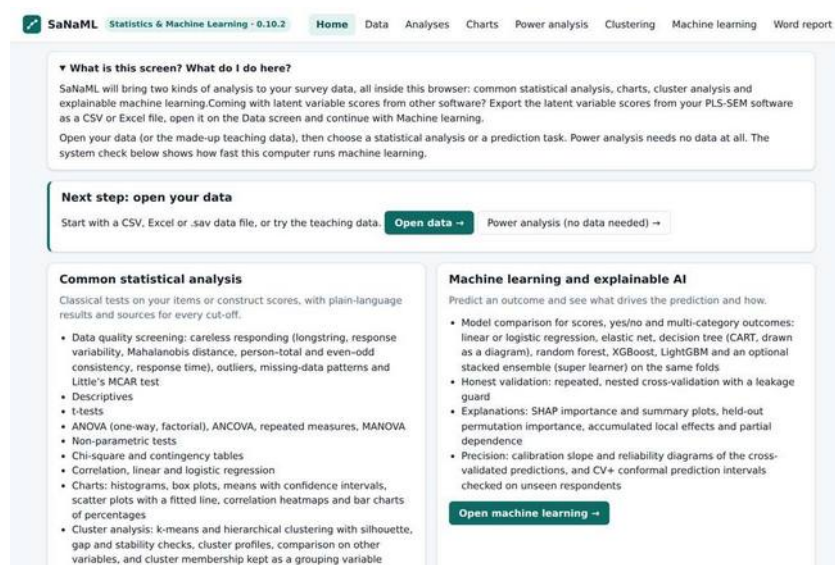


Figure 1. The Home screen of SaNaML.

Every screen starts with a box *What is this screen? What do I do here?*, which explains the screen in two or three sentences. Screens that need data show "No data yet" with a **Go to Data →** button until you open a file. SaNaML keeps your data, results and report list in this browser and restores them on your next visit; they are lost if you clear the browser's site data, so download the Word report (chapter 13) to keep your results.

Note: what this manual covers

This manual covers the common statistical tests and the way to prepare data for them. SaNaML also has screens for clustering and a full machine learning comparison; this manual uses machine learning only as a short check on a regression (chapter 12). SaNaML has no factor analysis screen: scale scores are built from items you choose, with reliability checked by Cronbach's α (Section 1.5).

1.2 Open the practice file

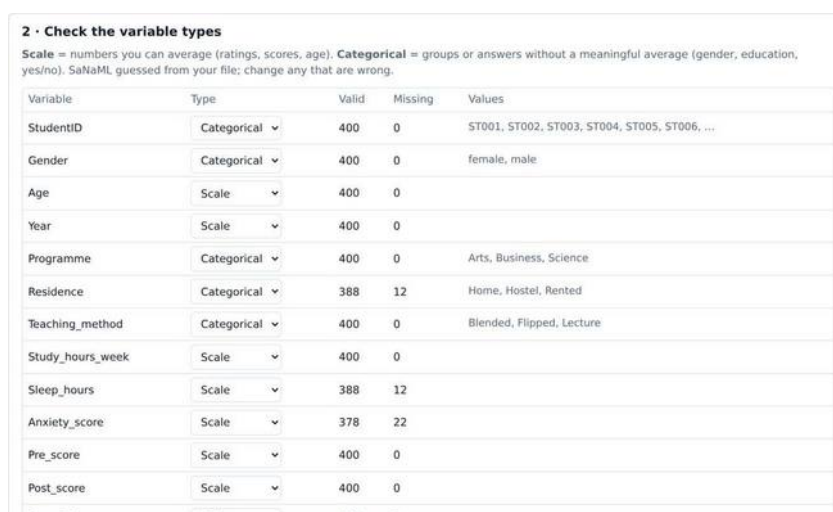
1. Click **Data** in the menu.
2. Click **Choose a file...** and select `practice_students.csv`. SaNaML reads the file in the browser.
3. SaNaML reports "Open `practice_students.csv`: 400 rows, 33 variables." **Close this data** closes the file again.

1.3 Check the variable types

The second card, 2 · *Check the variable types*, lists every variable with its type, the number of valid and missing values, and the categories of each categorical variable (Figure 2). SaNaML guesses the type from the file:

- **Scale:** numbers you can average (scores, hours, ratings).
- **Categorical:** groups or answers without a meaningful average (gender, programme, yes/no).

The type decides which analyses offer the variable: scale variables appear as outcomes and correlation variables, categorical ones as groups, factors and chi-square variables. Change any type that is wrong in its list.



2 · Check the variable types

Scale = numbers you can average (ratings, scores, age). Categorical = groups or answers without a meaningful average (gender, education, yes/no). SaNaML guessed from your file; change any that are wrong.

Variable	Type	Valid	Missing	Values
StudentID	Categorical	400	0	ST001, ST002, ST003, ST004, ST005, ST006, ...
Gender	Categorical	400	0	female, male
Age	Scale	400	0	
Year	Scale	400	0	
Programme	Categorical	400	0	Arts, Business, Science
Residence	Categorical	388	12	Home, Hostel, Rented
Teaching_method	Categorical	400	0	Blended, Flipped, Lecture
Study_hours_week	Scale	400	0	
Sleep_hours	Scale	388	12	
Anxiety_score	Scale	378	22	
Pre_score	Scale	400	0	
Post_score	Scale	400	0	

Figure 2. Step 2: the variable types of *practice_students.csv*.

What it means for our example

Gender (female, male), Programme (Arts, Business, Science), Residence (Home, Hostel, Rented), Teaching_method (Blended, Flipped, Lecture), Passed, Pass_before and Pass_after (No, Yes) are categorical. Everything else is scale, including Age, Year (1 to 4), the three repeated scores and the items Mot1 to Anx6. StudentID is listed as a categorical variable with 400 groups: it is only a label, so do not use it in an analysis.

Some variables can be treated either way. Year has only four values, 1 to 4. Treated as scale, it can be correlated or used as a covariate; treated as categorical, it can define groups (Figure 3). Choose according to the question, not the number of values.



Age	Scale	400	0	
Year	Categorical	400	0	1, 2, 3, 4
Programme	Categorical	400	0	Arts, Business, Science

Figure 3. Year changed from scale to categorical: SaNaML now lists its categories 1, 2, 3, 4.

In the practice file, change Year back to scale before you continue.

1.4 Missing data

Blank cells are read as missing values, and the *Missing* column counts them. In *practice_students.csv* Residence (12), Sleep_hours (12), Anxiety_score (22), Exam_score (13) and Satisfaction_1to5 (16) have missing values, between 3% and 6% of the 400 students.

Concept: how SaNaML treats missing values

SaNaML does not fill in missing values for the classical tests. Each analysis uses the cases that have the values it needs, and tells you how many that was:

- A t-test, ANOVA or descriptive table uses every case with a value on the variables involved (for example $n = 387$ for Exam_score).
- A correlation matrix uses, for each pair, every case with both values; a partial correlation uses complete cases only.
- A regression, repeated-measures ANOVA, factorial ANOVA or MANOVA uses only complete cases, and says so ("46 respondents with missing values left out").
- The machine learning screen fills in missing values inside each training set (chapter 12).

Under about 10% missing per variable is usually harmless if the values are missing at random (Hair et al., 2019). The data quality screen of chapter 2 adds Little's test of whether the missing values are missing completely at random (Little, 1988). A missing value is different from a zero: never type 0 for "no answer".

1.5 Scale scores and reliability

Many questionnaires measure one idea with several items. *practice_students.csv* has two such sets: Mot1 to Mot6 (motivation) and Anx1 to Anx6 (anxiety), each answered from 1 to 5. A **construct score** (scale score) is the mean of the items, which is more reliable than any single item. SaNaML builds it on the **Data** screen in the card 3 · *Build construct scores (optional)*, and first checks that the items hang together.

Concept: Cronbach's alpha and reverse-coded items

Cronbach's α estimates how consistently the items measure the same thing (Cronbach, 1951). It rises when the items correlate and when there are more of them. The **corrected item-total correlation** is the correlation of one item with the mean of the others. If one item is worded in the opposite direction ("I feel calm"), its correlation is negative until you **reverse-code** it: on a 1 to 5 scale a 2 becomes 4 (the lowest plus the highest answer, minus the value).

Build Motivation

1. In **Name of the construct**, type **Motivation**.
2. In **Quick pick: items starting with**, type **Mot** and press Tab. The items Mot1 to Mot6 are ticked and **Lowest answer** and **Highest answer** are filled in from the data (1 and 5).
3. Read Cronbach's α and the item statistics.

- Click **Create the construct score**. Motivation appears in the variable list with a "construct" label.

What it means for our example

Motivation has $\alpha = .86$ on 400 complete responses. The corrected item-total correlations are between .55 (Mot5) and .72 (Mot1), and deleting an item would never raise α above .86 (deleting Mot5 would leave it at .86): the six items form a consistent scale.

Build Anxiety, with two reversed items

In this file Anx3 and Anx5 are worded in the opposite direction and must be reversed.

- In **Name of the construct**, type **Anxiety** (the name Anxiety_score is already used by another variable).
- In **Quick pick**, type **Anx** and press Tab. The quick pick matches every variable that starts with "Anx", so it also ticks **Anxiety_score**. Untick Anxiety_score.
- Before reversing anything, look at the result.
- Tick the **R** box next to Anx3 and Anx5 (Figure 4).
- Click **Create the construct score**.

3 • Build construct scores (optional)

A construct score is the average of the items that measure one idea (for example PE1-PE4 → PE). SaNaML first checks the items hang together (Cronbach's α). If you already have latent variable scores from other software, open that file instead; it needs no construct building.

Name of the construct:

Letters, numbers and underscores: must not be used by another variable yet.

Quick pick: items starting with

Lowest answer: Highest answer:

Used to reverse-code items: on a 1-7 scale, a reversed 2 becomes 6.

Items (tick to include; "R" = reverse-coded)

☐ Mot3	☐ R
☐ Mot4	☐ R
☐ Mot5	☐ R
☐ Mot6	☐ R
☒ Anx1	☐ R
☒ Anx2	☐ R
☒ Anx3	☒ R
☒ Anx4	☐ R
☒ Anx5	☒ R
☒ Anx6	☐ R

$\alpha = .88$ Cronbach's α on 400 complete responses. Cronbach's α of .70 or higher is usually acceptable for research scales (Nunnally (1978)).

Item	Corrected item-total r	α if deleted
Anx1	.74	.86
Anx2	.69	.86
Anx3 (R)	.70	.86
Anx4	.66	.87
Anx5 (R)	.72	.86
Anx6	.66	.87

An item-total correlation below .30 (or a negative one) suggests the item does not fit, or that it needs reverse-coding.

Create the construct score

Figure 4. Anxiety with Anx3 and Anx5 reverse-coded.

What it means for our example

Without reversal, Anx3 and Anx5 have corrected item-total correlations of $-.58$ and $-.61$, and $\alpha = -.25$: a negative α is a clear sign that items point in different directions. With Anx3 and Anx5 reversed, $\alpha = .88$ and every corrected item-total correlation lies between .66 and .74. Always read the item statistics before you create a score.

Cut-off values: internal consistency

Cronbach's α of .70 or higher is usually acceptable for research scales (Nunnally, 1978). The pill is green at .70 or more, amber from .60 and red below.

Corrected item-total correlation below .30, or negative, suggests that the item does not fit or needs reverse-coding.

How to report it

"Motivation (six items, 1 to 5) was reliable, Cronbach's α = .86, and so was Anxiety (six items, two reverse-coded), α = .88."

1.6 Compute and recode variables

Card 4 · *Compute and recode variables (optional)* on the Data screen makes a new variable from an existing one. The original is never changed: the new variable is added to the list with a name made from the original and a suffix (Anxiety_score_In), and every analysis offers it like any other variable. Choose what you want to make in **What do you want to make?:**

- **Transform a scale variable:** the natural logarithm $\ln(x)$, $\ln(1 + x)$, the square root, the square, a standardised score (z-score), the centred score (mean subtracted) or a reverse-coded score (lowest + highest – x). The logarithm needs $x > 0$ and the square root $x \geq 0$; values that cannot be transformed become missing, and SaNaML says how many respondents have a value.
- **Make groups from a scale variable:** two groups at the median, three (tertiles) or four (quartiles) equal-sized groups, or your own cut points. A value equal to a cut point goes to the lower group.
- **Make a 0/1 indicator of a category:** 1 for the ticked categories and 0 for all others.
- **Combine two scale variables:** the difference $A - B$, the sum, the product, the ratio $A \div B$ or the percentage change $(A - B) \div B \times 100$. A missing value in either variable, or a zero denominator, gives a missing result.
- **Merge categories into one:** two or more categories become a single category with a name you give.

A logarithm of Anxiety_score

1. Choose **Transform a scale variable (logarithm, standardise, reverse ...)**.
2. In **Variable** choose **Anxiety_score** and in **Transformation** choose **Natural logarithm, $\ln(x)$** (Figure 5).
3. Leave the suggested name and click **Add the new variable**. SaNaML reports: "Added Anxiety_score_In: 378 of 400 respondents have a value." The 22 students with a missing anxiety score keep a missing value.

4 • Compute and recode variables (optional)
 Make a new variable from an existing one: a logarithm for a skewed variable, a standardised score, low and high groups, a 0/1 indicator for one category, or merged categories. The original variable is never changed.

What do you want to make?
 Transform a scale variable (logarithm, standardise, reverse ...)

Variable
 Anxiety_score

Transformation
 Natural logarithm, ln(x)

Name of the new variable
 Anxiety_score_ln

Letters, numbers, dots and underscores; must not be used by another variable yet.

Add the new variable

Pulls in a long right tail (income, salary, spend). Needs $x > 0$; other values become missing.

Figure 5. Card 4 with Anxiety_score and the natural logarithm chosen.

Now describe both variables (**Analyses, Descriptive statistics**). The skewness of Anxiety_score is 0.97 (kurtosis 1.74); that of Anxiety_score_ln is 0.26 (kurtosis 0.20), with $M = 2.58$, $SD = 0.21$ and $n = 378$ (Figure 6). The logarithm pulled in the long right tail. Results for the transformed variable are on the logarithmic scale: report them as such, and take the logarithm only when the distribution or the research question calls for it.

What it means: No variable is strongly skewed (all $|skewness| \leq 1$).

Descriptive statistics									
Variable	n	Missing	Mean	SD	Median	Min	Max	Skewness	Kurtosis
Anxiety_score	378	22	13.50	2.90	13.10	7.10	26.20	0.97 (0.13)	1.74 (0.25)
Anxiety_score_ln	378	22	2.58	0.21	2.57	1.96	3.27	0.26 (0.13)	0.20 (0.25)
Passed_Yes	400	0	0.69	0.46	1.00	0.00	1.00	-0.85 (0.12)	-1.28 (0.24)

Skewness and kurtosis are bias-corrected (G1, G2; kurtosis is excess kurtosis); standard errors in brackets. Median = weighted average at $(n + 1)p$.

Figure 6. Descriptive statistics of Anxiety_score, its logarithm and the indicator Passed_Yes.

Groups from a scale variable

Choose **Make groups from a scale variable (median split, tertiles ...)**, **Study_hours_week** and **Two groups at the median**. SaNaML shows the cut point (6.3 hours) before you add the variable; the new variable Study_hours_week_group has the groups "Low ($\leq$ median)" and "High ($>$ median)".

Compare the exam scores of the two groups with an independent-samples t-test (Chapter 5): Low $M = 51.36$ ($SD = 8.69$, $n = 195$), High $M = 58.63$ ($SD = 9.55$, $n = 192$), Welch's $t(380.42) = -7.82$, $p < .001$, $d = -0.80$, 95% CI $[-1.00, -0.59]$; the negative sign only means that the first group (Low) scored lower. The same groups are related to passing: 56% of the 202 low-hours students passed against 83% of the 198 high-hours students, $\chi^2(1, N = 400) = 35.40$, $p < .001$, Cramér's $V = .30$.

Note: think before you split a scale variable

Splitting a scale variable throws information away and the result depends on where the cut point falls. Use groups when the question is about groups (heavy against light users, tertiles of a score) and keep the variable as it is when you can: the correlation of Study_hours_week with Exam_score in chapter 10 uses every student's exact hours.

An indicator and merged categories

Choose **Make a 0/1 indicator of a category**, **Passed** and tick **Yes**: Passed_Yes is 1 for the students who passed and 0 for the others (400 of 400 have a value). Its mean is the proportion who passed: $M = 0.69$ (278 of 400 students). Choose **Merge categories into one**, **Residence**, tick **Hostel** and **Rented**, and call the merged category *Away from home*. Residence_merged has the categories Home and

Away from home; the 12 students with a missing residence stay missing (388 of 400). Exam_score does not differ between them: Home $M = 54.83$ ($SD = 9.58$, $n = 189$) and Away from home $M = 55.24$ ($SD = 10.04$, $n = 186$), Welch's $t(371.49) = -0.41$, $p = .682$, $d = -0.04$, 95% CI $[-0.24, 0.16]$.

A gain score from two variables

Choose **Combine two scale variables (difference, change, ratio ...)**, **Post_score** as **Variable A**, **Difference, A – B** and **Pre_score** as **Variable B** (Figure 7). Call the new variable Gain and add it: "Added Gain: 400 of 400 respondents have a value." The one-way ANOVA of this gain by teaching method is in Section 6.5.

4 • Compute and recode variables (optional)

Make a new variable from an existing one: a logarithm for a skewed variable, a standardised score, low and high groups, a 0/1 indicator for one category, or merged categories. The original variable is never changed.

What do you want to make?
 Combine two scale variables (difference, change, ratio ...)

Variable A
 Post_score

How to combine A (the variable above) and B
 Difference, A – B

Variable B
 Pre_score

Name of the new variable
 Gain

Letters, numbers, dots and underscores; must not be used by another variable yet.

Add the new variable

A change or gain score: after minus before.

Figure 7. Combining *Post_score* and *Pre_score* into the gain.

Exercise: scales in *practice_brand_survey.csv* and *practice_engagement.csv*

Open *practice_engagement.csv* and build a scale for the three Vigor items (Vigor1 to Vigor3; they are rated 1 to 7, so check the lowest and highest answer). Do the same for Dedication and Absorption. Which scale has the highest α ? Then open *practice_brand_survey.csv* and build Trust (Trust1 to Trust4), Value (Value1 to Value4) and Loyalty (Loyalty1 to Loyalty3). None of these items needs reversing. Answers: Vigor $\alpha = .83$, Dedication $\alpha = .84$ and Absorption $\alpha = .79$ (so Dedication is the most consistent of the three); Trust $\alpha = .87$, Value $\alpha = .81$ and Loyalty $\alpha = .84$.

2. Checking data quality

Before you test anything, look for respondents who answered carelessly, for extreme values and for patterns of missing data. SaNaML's data quality screen does this and suggests whom to exclude; you decide, and nothing is deleted from your file.

Concept: careless responding

Some respondents answer without reading: the same answer everywhere (straight-lining), random answers, or far too fast. Their answers weaken correlations and can create or hide effects. No single index catches every kind of careless answering, so the usual advice is to compute several indices and exclude only respondents flagged by more than one (Curran, 2016; Meade & Craig, 2012).

The practice file contains eight students who answered every item with 3. Let us find them.

2.1 The screening settings

1. Build the Motivation and Anxiety scores first (Section 1.5). SaNaML takes the questionnaire items, the scales and the reverse-coded items from the construct scores.
2. On **Data**, click **Check data quality** → (or choose **#/quality** from the card at the bottom of the Data screen).
3. Check the card *Screening settings (What to check)* (Figure 8).
4. Click **Screen the data**.

Screening settings (What to check)

Questionnaire items (in the order of the file)

Select all Select none

☐ Age ☐ Year ☐ Satisfaction_1to5 ☒ Mot1 ☒ Mot2 ☒ Mot3 ☒ Mot4

☒ Mot5 ☒ Mot6 ☒ Anx1 ☒ Anx2 ☒ Anx3 ☒ Anx4 ☒ Anx5

☒ Anx6

Completion time in seconds -- none -- Suggest exclusion when flagged by 2 or more indices (Recommended) Longstring flag at half the its

Reverse-coded items are taken from the construct scores you built; scales come from the constructs, the PLS-SEM model or item names (PE1, PE2 ... form PE).

Screen the data

Figure 8. The screening settings: the twelve questionnaire items are ticked.

The list *Questionnaire items (in the order of the file)* offers every variable with whole-number answers and few values. With the constructs built, the twelve items Mot1 to Anx6 are ticked and Age, Year and Satisfaction_1to5 are not. If you have not built constructs, SaNaML ticks all the candidates: untick everything that is not an item of the questionnaire block you want to screen, because the indices compare a person's answers across the items you tick. **Completion time in seconds** is left at none (the practice file has no timing variable), **Suggest exclusion when flagged by** stays at **2 or more indices (Recommended)**, and **Longstring flag at** is left empty, which means half the items (here 6 of 12).

2.2 Reading the screening report

The report checks four things: careless responding, outliers ($|z| > 3.29$ on the questionnaire items, the construct scores and the other numeric variables of the file, and multivariate outliers on the construct scores by Mahalanobis D^2 with $p < .001$; Tabachnick & Fidell, 2019), the amount of missing data, and, when there are missing values among the items, Little's MCAR test (Little, 1988).

Table 3

The careless-response indices

Index	Flag when
Longstring (same answer many times in a row)	as many identical answers in a row as the longstring flag (here 6 of 12)
Intra-individual response variability	the standard deviation of the person's answers is 0
Mahalanobis distance (unusual answer pattern)	D^2 with $p < .001$
Person-total correlation (answers unlike everyone else's)	$r < 0$; used only when the item means differ enough (SD of the item means at least 0.25)
Even-odd consistency (inconsistent within scales)	needs at least four scales with two or more items
Response time (too fast to have read the items)	needs a completion-time variable

What it means for our example

Three indices were computed on 12 items; the person-total correlation was not used because the item means are too similar (SD 0.04), the even-odd index needs at least four scales and we have two, and there is no completion time. Eight respondents (2.0% of 400) were flagged by two or more indices and are suggested for exclusion. Longstring flagged 28 respondents (7.0%), intra-individual response variability (an SD of zero) flagged 8 (2.0%), and Mahalanobis distance flagged none. The 20 respondents flagged only by longstring have runs of 6 to 9 identical answers but vary elsewhere (their response variability is above zero); this is why one index alone is not enough. Eight values are beyond $|z| = 3.29$, in two variables (Section 2.3), and nobody is a multivariate outlier on the construct scores.

Respondents suggested for exclusion						
Respondent	Flags	Longstring	IRV	D^2 (p)	Person-total r	Even-odd r
ST072	Longstring, Intra-individual response variability	12	0.00	0.0 (1.000)	-	-
ST109	Longstring, Intra-individual response variability	12	0.00	0.0 (1.000)	-	-
ST119	Longstring, Intra-individual response variability	12	0.00	0.0 (1.000)	-	-
ST172	Longstring, Intra-individual response variability	12	0.00	0.0 (1.000)	-	-
ST278	Longstring, Intra-individual response variability	12	0.00	0.0 (1.000)	-	-
ST293	Longstring, Intra-individual response variability	12	0.00	0.0 (1.000)	-	-
ST299	Longstring, Intra-individual response variability	12	0.00	0.0 (1.000)	-	-
ST315	Longstring, Intra-individual response variability	12	0.00	0.0 (1.000)	-	-

Figure 9. The respondents suggested for exclusion: all eight have a longstring of 12 and no variation.

[fig:q-outliers|The outliers table: values beyond $|z| = 3.29$ in numeric variables, with the respondents. |420]

The eight respondents (Figure 9) answered all 12 items with the same value: the students ST072, ST109, ST119, ST172, ST278, ST293, ST299 and ST315. SaNaML labels respondents by the identifier

column of the file, here StudentID (it recognises a text identifier such as ST072 by itself); a file without such a column is labelled by row number. The dashed line in the longstring histogram marks the flag.

2.3 Outliers and missing values

The screening also looks for outliers in every numeric variable of the file, not only in the questionnaire items. The *Outliers* table (Figure 10) lists, for each variable, the number of values beyond $|z| = 3.29$, the largest $|z|$ and the respondents concerned. Study_hours_week has five (largest $|z| = 9.09$: ST037, ST114, ST122, ST236 and ST333; up to 60 hours a week against a median of 6.30) and Anxiety_score has three (largest $|z| = 4.38$: ST103, ST172 and ST286). The check lists them; it does not remove them. An outlier is not an error by itself: check it against the questionnaire, keep genuine extreme answers, correct typing errors and report what you did. A box plot (Section 3.4) shows these extreme study times as well. Note that ST172 is both an outlier on Anxiety_score and one of the eight straight-liners.

To see the missing-data checks at work, tick Satisfaction_1to5 as a thirteenth item and screen again. SaNaML then reports that 0.3% of the answers are missing (at most 4.0% for one item), that 384 respondents answered every item, and that Little's test does not reject missing completely at random, $\chi^2(12) = 6.18$, $p = .907$. Only two of the eight straight-liners are then suggested for exclusion (ST072 and ST299, whose satisfaction rating is also 3): for the other six the run of identical answers ends at the thirteenth item and their response variability is no longer zero. The lesson is to screen the items of one questionnaire block, not unrelated variables. Untick Satisfaction_1to5 and screen again before you continue.

Cut-off values: missing data

Under 10% missing per variable is usually ignorable if the missingness is random; more needs a principled method such as FIML or multiple imputation (Hair et al., 2019).

Little's MCAR test $p \geq .05$: no evidence against missing completely at random (Little, 1988). A non-significant result does not prove it, and a significant one does not rule out data that are missing at random.

2.4 Your decision

The card *Your decision (Which respondents to exclude)* lists every respondent with at least one flag, 28 here; those suggested by the rule are ticked (Figure 11).

1. Look at the flagged respondents, and untick anyone you want to keep (or tick others).
2. Click **Add to Word report** under the screening report, so that the rule, the cut-offs and the number excluded are documented.
3. Click **Exclude the 8 ticked respondents**. The card *Screened data in use* confirms the exclusion: every analysis now uses the remaining 392 students.

Your decision (Which respondents to exclude)

Respondents with at least one flag (28); ticked = to be excluded

Exclude	Respondent	Flags	Longstring	IRV
☐	ST005	Longstring	9	0.58
☐	ST018	Longstring	9	0.43
☐	ST028	Longstring	7	0.51
☐	ST031	Longstring	6	0.39
☒	ST072	Longstring, Intra-individual response variability	12	0.00
☒	ST109	Longstring, Intra-individual response variability	12	0.00
☒	ST119	Longstring, Intra-individual response variability	12	0.00
☐	ST148	Longstring	6	0.60
☐	ST170	Longstring	6	0.45
☒	ST172	Longstring, Intra-individual response variability	12	0.00

Exclude the 8 ticked respondents

Figure 11. Your decision: the flagged respondents, with the eight suggested for exclusion ticked.

Undo the exclusion (use all 400 respondents) brings everyone back.

Note: the rest of this manual uses all 400 students

The eight straight-liners answered only the Mot and Anx items; their other variables (exam scores, hours, groups) are probably fine, and excluding them would change every number below. To let you reproduce the numbers in this manual, undo the exclusion before you continue. In your own study you would keep the exclusion, report it, and run every analysis on the screened data.

How to report it

"Data were screened for careless responding with three indices (longstring, intra-individual response variability, Mahalanobis distance; Curran, 2016; Meade & Craig, 2012). Eight of 400 respondents were flagged by at least two indices (each had answered all 12 items identically) and were excluded. Eight values on two variables lay beyond $|z| = 3.29$; they were checked and kept."

Exercise: find the straight-liners in practice_brand_survey.csv and careless_demo.csv

The brand survey has ten respondents who answered every item with 5. Open practice_brand_survey.csv, build Trust, Value and Loyalty scores, screen the data and check that ten respondents are suggested. Then open careless_demo.csv (300 respondents, 20 items and a Duration variable): choose Duration as **Completion time in seconds** and compare the number of respondents suggested when you ask for 1, 2 or 3 flags. Which setting would you report, and why? Answers: in the brand survey ten respondents are suggested (2.0% of 500; the even-odd index is not used because it needs at least four scales and there are three). In careless_demo.csv, five indices are computed on 20 items; 1 flag suggests 40 respondents (13.3%), 2 flags suggest 13 (4.3%) and 3 flags suggest nobody. The screen also lists three outliers on Duration (P011, P012 and P013; largest $|z| = 4.01$).

3. Describing your data

Describe your data before you test anything. Descriptive statistics tell you the typical values and the spread; charts show shape, unusual values and relationships that a single number hides. Both are on the [Analyses](#) and [Charts](#) screens.

3.1 Descriptive statistics

Click [Analyses](#) in the menu. The list on the left shows the analyses, each with a one-line description; the result appears on the right. Choose [Descriptive statistics](#), tick the scale variables to describe (here `Study_hours_week`, `Sleep_hours`, `Anxiety_score` and `Exam_score`) and, optionally, a categorical variable in [Split by](#).

There is **no Run button** anywhere in the Analyses: the result appears as soon as the variables are chosen, and changes when you change them. Every result has the same parts:

- a title, for example *Descriptive statistics by Gender*;
- **What it means**: the finding in one or two plain sentences;
- the **checks**, each with a verdict (OK, Note, Check or Info), an explanation and the rule with its source;
- the **tables** (and figures);
- **How to report it (APA style)**: a ready-made sentence, with [Copy this sentence](#) and [Add to Word report](#);
- **References**, collapsed at the bottom.

The table gives the number of valid values (n), the number missing, the mean, standard deviation (SD), median, minimum, maximum, and bias-corrected skewness and kurtosis with their standard errors in brackets.

Concept: mean, median, skewness and kurtosis

The **mean** is the arithmetic average; the **median** is the middle value. In a symmetric distribution they are close. A few very large values pull the mean up but hardly move the median, so a mean well above the median signals a right-skewed distribution. **Skewness** measures this asymmetry (0 = symmetric; positive = long tail to the right). **Kurtosis** here is excess kurtosis (0 = like a normal curve; large positive values = heavy tails or extreme values). SaNaML flags a variable when $| \text{skewness} | > 1$.

What it means for our example

`Study_hours_week` has $M = 7.28$ and $SD = 5.80$ but a median of only 6.30, a maximum of 60.00 hours, skewness 5.63 and excess kurtosis 41.62: a few extreme values dominate its shape. SaNaML therefore says that 1 of the 4 rows shows clear skewness and advises reporting medians and reading the rank-based tests. `Anxiety_score` is right-skewed too (skewness 0.97, $M = 13.50$, $Mdn = 13.10$, $SD = 2.90$, $n = 378$ with 22 missing), just under the flag. `Sleep_hours` ($M = 6.68$, $SD = 1.07$, $n = 388$) and `Exam_score` ($M = 54.97$, $SD = 9.81$, $n = 387$) are close to symmetric.

Choose **Gender** in **Split by** to see each variable for women and men (Figure 12). The mean Exam_score is 56.57 (SD = 9.64, n = 239) for women and 52.38 (SD = 9.57, n = 148) for men: a difference of about four points that chapter 5 will test. The APA sentence under the table now lists each group (for example "Exam_score: Gender = female, M = 56.57, SD = 9.64, n = 239; Gender = male, M = 52.38, SD = 9.57, n = 148"). Note that a descriptive table only describes; it does not say whether a difference could be due to chance.

What it means: 3 of the 8 rows show clear skewness ($|skewness| > 1$): report medians as well, and read the rank-based tests next to the tests of means.

Descriptive statistics										
Variable	Gender	n	Missing	Mean	SD	Median	Min	Max	Skewness	Kurtosis
Study_hours_week	female	246	0	7.47	6.35	6.40	1.40	60.00	5.40 (0.16)	36.38 (0.3)
Study_hours_week	male	154	0	6.96	4.79	6.20	0.70	52.50	5.87 (0.20)	53.24 (0.3)
Sleep_hours	female	238	0	6.71	1.03	6.70	3.90	9.90	0.08 (0.16)	0.29 (0.3)
Sleep_hours	male	150	0	6.61	1.14	6.70	3.50	9.80	-0.13 (0.20)	0.18 (0.3)
Anxiety_score	female	233	0	13.30	2.75	13.00	7.10	26.20	1.04 (0.16)	2.75 (0.3)
Anxiety_score	male	145	0	13.81	3.11	13.20	8.40	24.50	0.85 (0.20)	0.68 (0.4)
Exam_score	female	239	0	56.57	9.64	55.70	33.10	80.10	0.07 (0.16)	-0.24 (0.3)
Exam_score	male	148	0	52.38	9.57	51.35	29.00	75.10	0.13 (0.20)	-0.35 (0.3)

Skewness and kurtosis are bias-corrected (G1, G2; kurtosis is excess kurtosis); standard errors in brackets. Median = weighted average at $(n + 1)p$.

Figure 12. The same variables split by Gender.

Note: the sentence under a split table

The *How to report it* sentence of the descriptive table lists the overall mean, SD and n of each variable; it does not repeat the group values. When you split by a variable, copy the group rows from the table instead.

3.2 The Charts screen

Click **Charts** in the menu. The card 1 · Choose the chart has a **Chart type** list and the variables the chart needs; the chart appears below with a short description, a summary table and a caption you can add to the Word report. Six chart types are available: histogram, box plot, means with confidence intervals, scatter plot with a fitted line, correlation heatmap and bar chart of percentages.

Note: charts show patterns, not proof

An apparent difference in a chart needs a test, and a pattern in a scatter plot can come from a few unusual respondents.

3.3 Histogram: the shape of one variable

Choose **Histogram (How are the values spread?)** and Anxiety_score as the **Variable** (Figure 13). Each bar counts the respondents whose value falls in a band; the lines mark the mean and the median.

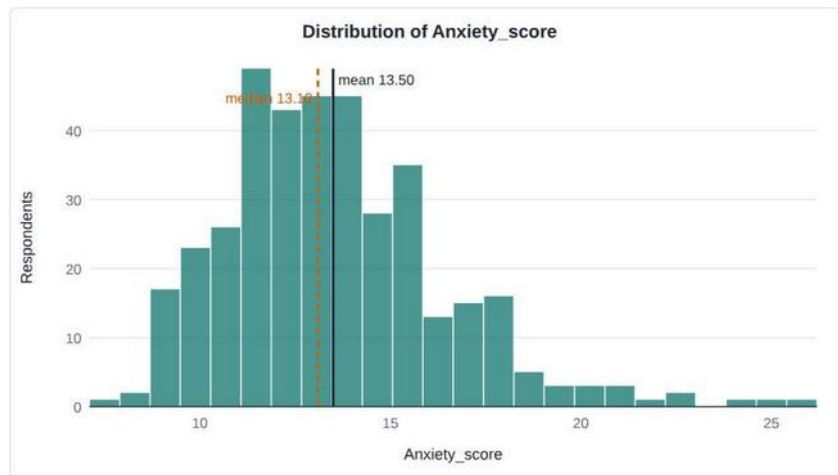


Figure 13. Histogram of *Anxiety_score* with the mean (13.50) and median (13.10).

SaNaML summarises it: *Anxiety_score* has 378 values with mean 13.50, median 13.10 and SD 2.90, and the distribution is skewed to the right (a long tail of high values). Many tests of means tolerate this with 30 or more cases per group; for a skewed outcome in small groups, prefer the rank-based test.

3.4 Box plot: medians, spread and unusual values

Choose **Box plot (Medians and spread, by group)** and *Study_hours_week* (Figure 14). The box spans the middle half of the values (the quartiles), the line is the median, the whiskers reach the most extreme values within 1.5 times the interquartile range, and the circles are unusual values beyond that.

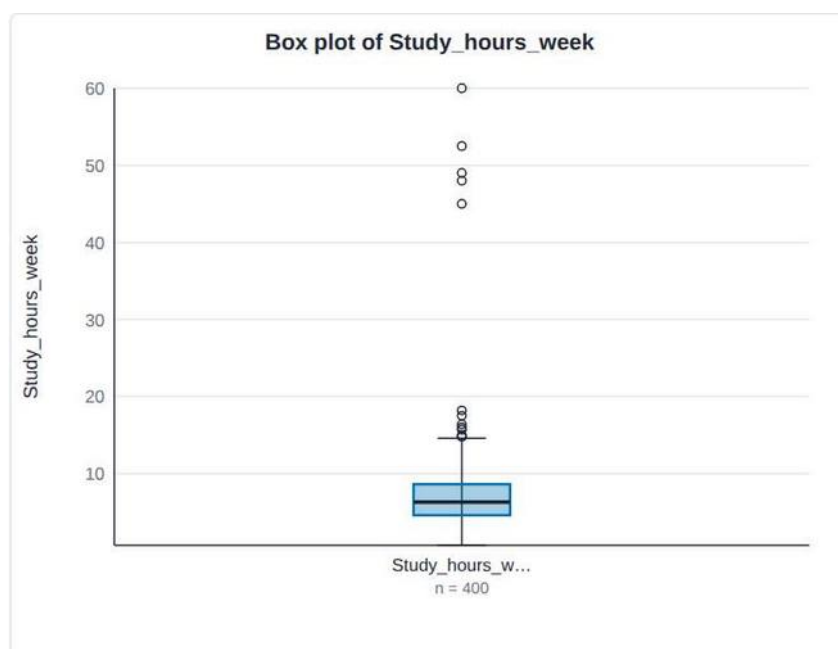


Figure 14. Box plot of *Study_hours_week*: the circles are unusual values.

The median is 6.30 hours and the quartiles are 4.60 and 8.70. Five students (with 45 to 60 hours a week) lie far above the rest, and several more lie above the whisker at about 15 hours. Before you analyse such a variable, ask whether the extreme values are errors (a typing mistake, hours per month instead of per week), a different kind of student, or a true feature of the data. Do not delete

them just because they are inconvenient: compare the result with and without them, and use rank-based methods (Section 10.2), which are not pulled by single extreme values.

Choose Exam_score and **Groups (optional)** Gender to compare groups. The medians are 55.70 for women and 51.35 for men.

3.5 Means with confidence intervals

Choose **Means with confidence intervals (Group averages)**, Exam_score and **Groups** Teaching_method (Figure 15).

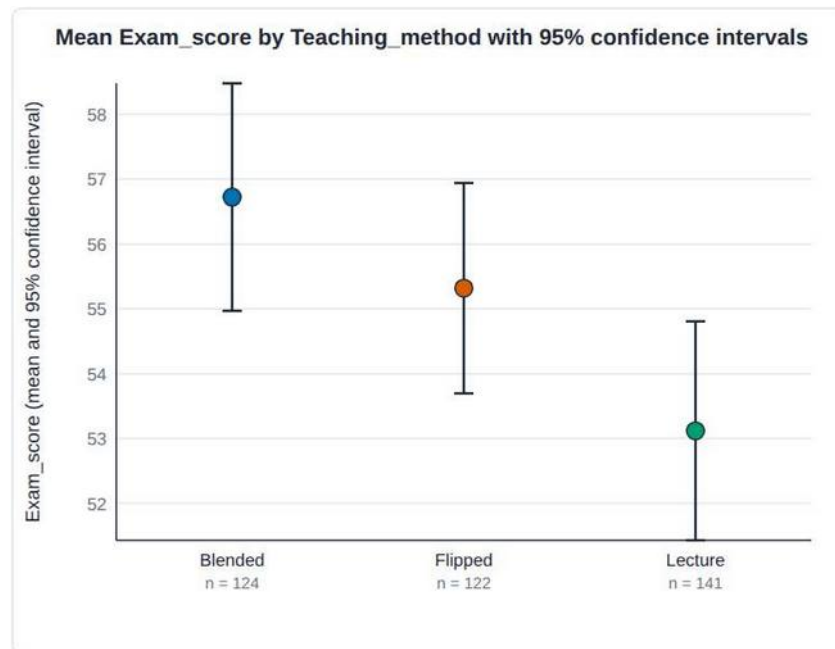


Figure 15. Mean Exam_score by Teaching_method with 95% confidence intervals.

Concept: confidence interval

A 95% confidence interval is a range computed so that, in repeated samples, 95% of such intervals would contain the true population mean. It shows how precisely each group mean is estimated. Where two intervals do not overlap, the groups very likely differ; overlapping intervals do not by themselves show that they are the same. Run the comparison in the Analyses to test it.

The means are Blended 56.72 [54.97, 58.48], Flipped 55.32 [53.70, 56.94] and Lecture 53.12 [51.43, 54.81]. The intervals of Blended and Lecture hardly separate; chapter 6 tests this properly.

3.6 Scatter plot with a fitted line

Choose **Scatter plot with line (Do two variables go together?)**, Study_hours_week on the **Horizontal axis** and Exam_score on the **Vertical axis** (Figure 16). Optionally choose **Groups (optional)** to colour the dots by group.

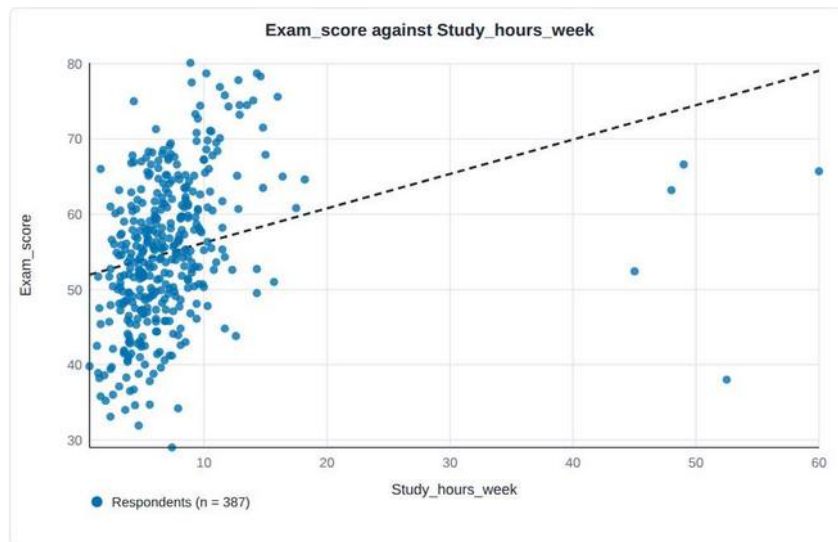


Figure 16. *Exam_score against Study_hours_week with the least-squares line.*

The relationship is small and positive (Pearson $r = .27$, $n = 387$). The dashed line is the straight-line fit: Exam_score changes by 0.457 for each one-unit increase in Study_hours_week (intercept 51.632). Look at the dots before you trust r : a few extreme study hours on the right pull the line, and the cloud may curve rather than follow a straight line. Chapter 10 compares Pearson's r with rank correlations on these data.

3.7 Bar chart of percentages

Choose **Bar chart of percentages (Categories)** and Programme. The table gives counts and percentages.

Programme ($n = 400$): Business 40.8% (163 students), Science 36.3% (145) and Arts 23.0% (92). Add any chart to the Word report with **Add to Word report**; its caption sentence ("Figure. Distribution of ...") goes with it.

How to report it

"Weekly study time was strongly right-skewed ($M = 7.28$, $SD = 5.80$, $Mdn = 6.30$, skewness = 5.63), with a few students studying more than 40 hours; medians are therefore reported alongside means."

Exercise: charts on practice_marketing.csv and segments_demo.csv

Open practice_marketing.csv. Draw a histogram of Annual_spend and a box plot of Income_k: both are skewed (the file note says they were built that way), so a logarithm is often taken before a linear regression (Section 1.6). Draw a correlation heatmap of Income_k, Visits_per_month, Avg_basket and Annual_spend. Answer: Visits_per_month and Annual_spend correlate at $r = .87$, Income_k and Avg_basket at $r = .48$ and Avg_basket and Annual_spend at $r = .42$, while Income_k and Visits_per_month are unrelated ($r = -.00$). Then open segments_demo.csv and draw a scatter plot of Brand_loyalty against Price_sensitivity, grouped by Channel: the relationship is negative. Finally, take the natural logarithm of Annual_spend and of Income_k (Section 1.6) and describe them. Answer: the skewness of Income_k falls from 1.38 to 0.04; that of Annual_spend from 1.07 to -0.30, but its logarithm exists for only 1,418 customers, because the 82 customers with a spend of 0 become missing ($\ln(1 + x)$ would keep them).

4. Choosing the right test

Choosing a test is a decision about your data, not about the software. Four questions settle most cases.

1. **What kind of variable is the outcome?** A number or a rating you can average (scale), or a category (categorical)?
2. **How many groups or measurements are compared?** Two, or three or more?
3. **Are the measurements independent or paired?** Different people in each group (independent), or the same people measured more than once (paired or repeated)?
4. **What are you asking?** A difference between groups, a relationship between variables, or a prediction?

Concept: independent and paired data

Data are **independent** when each person contributes one value and what one person gives tells you nothing about another (men and women in a survey). Data are **paired** when the same people (or matched pairs) are measured again (a pre-test and a post-test). Ignoring the pairing throws away power; treating paired data as independent gives wrong p-values.

4.1 The Help me choose guide

SaNaML asks the first and the last question for you. Click [Analyses](#), then [Test selection guide \(Help me choose\)](#).

1. Answer 1 · *What kind of answer are you analysing?*: a continuous outcome (a number or a rating) or a categorical outcome (a category).
2. Answer 2 · *What do you want to compare or test?* (for a categorical outcome: *What do you want to know?*).
3. SaNaML shows the suggested test, for example **McNemar test**, and [Open it →](#) opens it with the right design already chosen.

Each test also computes its rank-based or exact alternative and tells you which one to read, based on your data, so the guide only needs to settle the design.

4.2 The decision table

Table 4

Which test for which question (the tests of SaNaML)

Outcome	Question and design	Test in SaNaML	Rank-based or exact alternative
Scale	Two independent groups	Independent-samples t-test (Welch)	Mann-Whitney U
Scale	Same people, two measurements	Paired-samples t-test	Wilcoxon signed-rank
Scale	One group against a fixed value	One-sample t-test	Wilcoxon signed-rank

Outcome	Question and design	Test in SaNaML	Rank-based or exact alternative
Scale	Three or more independent groups	One-way ANOVA (Welch, Tukey or Games-Howell)	Kruskal-Wallis with Dunn
Scale	Same people, three or more times	One-way repeated-measures ANOVA (Mauchly, Greenhouse-Geisser, Huynh-Feldt)	Friedman
Scale	Groups from two or more factors, or adjusted for a covariate	Factorial ANOVA, ANCOVA	none
Scale (several)	Several outcomes at once, one factor	One-way MANOVA	none
Categorical	Two categorical variables related?	Chi-square test of independence	Fisher's exact (2×2)
Categorical	Categories equally common?	Chi-square goodness-of-fit	none
Categorical	Same people, yes/no, two times	McNemar test	Exact binomial
Scale	How strongly do variables go together?	Pearson, Spearman, Kendall, partial correlation	Spearman or Kendall
Scale	Which variables predict a score?	Multiple linear regression (hierarchical)	none
Categorical (two)	Which variables predict a yes/no outcome?	Binary logistic regression	none

4.3 Parametric or rank-based?

Concept: assumptions of the tests of means

The t-test and ANOVA assume independent observations, a roughly normal distribution of the sample *mean* (which the central limit theorem delivers for moderate to large groups even when the data are skewed) and, for the classic versions, equal variances. SaNaML's default tests, Welch's t-test and Welch's ANOVA, do not need equal variances. The rank-based tests (Mann-Whitney, Wilcoxon, Kruskal-Wallis, Friedman, Spearman) need no normality, are not pulled by outliers, and are preferable for small samples of clearly skewed or ordinal data; with large samples both families usually agree. SaNaML reports both and tells you which one to read.

4.4 Multiple tests and error rates

Every test you run has a 5% chance of a false positive when there is no effect at $\alpha = .05$. Run twenty tests and you should expect one. Decide your tests before you look at the data, report all of them, and use the adjusted p-values SaNaML gives for families of comparisons: Tukey-Kramer or Games-Howell after an ANOVA, Dunn with the Holm adjustment after Kruskal-Wallis, Holm-adjusted

pairwise tests after repeated measures and factorial designs, and Holm-adjusted p-values in a correlation matrix (Holm, 1979).

Exercise: choose the test

For each question below, use the guide (or Table Table 4) to name the test: (a) Do students in the three programmes differ in Exam_score? (b) Did the same students' Pre_score rise by the time of Post_score? (c) Is the proportion who passed related to Gender? (d) Does Sleep_hours predict Exam_score? Answers: (a) One-way ANOVA; (b) paired-samples t-test; (c) chi-square test of independence; (d) correlation or multiple linear regression.

5. Comparing two groups or two measurements

This chapter covers the t-tests and their rank-based alternatives: two independent groups (Exam_score for women and men), the same students measured twice (Pre_score and Post_score) and one group against a fixed value.

Concept: the t-test and its effect size

A t-test divides the difference between two means by its standard error: the larger the ratio, the less likely a difference this large arises by chance if the true difference is zero. The p-value is that probability. It says nothing about how large or important the difference is. For that you need an **effect size**; for two means, Cohen's d: the difference in standard deviations (Cohen, 1988).

Always report both, with a confidence interval.

Choose **Independent-samples, paired-samples and one-sample t-tests (Compare two groups or two measurements)** in the Analyses menu. Under *What do you want to compare?* pick the design: **independent-samples, paired-samples** or **one-sample**. **Direction of the test** is two-tailed by default; choose a one-tailed test only if your hypothesis predicted the direction before you saw the data.

5.1 Independent-samples t-test: Exam_score by Gender

1. Choose **Independent-samples t-test (Two groups of different people)**.
2. In **Score to compare**, choose **Exam_score**; in **Groups**, choose **Gender** (Figure 17). If the grouping variable has more than two groups, two lists ask which two to compare; use One-way ANOVA (chapter 6) for all of them.

Independent-samples, paired-samples and one-sample t-tests (Compare two groups or two measurements)

What do you want to compare?

- ☒ **Independent-samples t-test (Two groups of different people)**
For example women vs men, or users vs non-users, on the same score.
- ☐ **Paired-samples t-test (The same people measured twice)**
For example satisfaction before and after, or two ratings by each respondent.
- ☐ **One-sample t-test (One score against a fixed value)**
For example: is the average agreement above the scale midpoint of 4?

Score to compare

Exam_score

A scale variable: an item, a construct score or another number.

Groups

Gender (2 groups)

A categorical variable. If it has more than two groups, choose the two to compare below (or use One-way ANOVA).

Direction of the test

- ☒ **Two-tailed test (Any difference)**
The usual choice: a difference in either direction counts.
- ☐ **Right-tailed (one-tailed) test (Only "first is greater")**

Figure 17. The t-test screen with the independent-samples design: Exam_score by Gender.

Group descriptives					
Group	n	Mean	SD	Median	Skewness
female	239	56.57	9.64	55.70	0.07
male	148	52.38	9.57	51.35	0.13

t-tests (two-sided)						
Test	t	df	p	Mean difference	95% CI	
Welch's t (recommended)	4.17	313.32	< .001	4.19	[2.21, 6.16]	
Student's t (equal variances)	4.17	385	< .001	4.19	[2.21, 6.16]	

Effect size			
Measure	Value	95% CI	Size
Cohen's d	0.44	[0.23, 0.64]	small
Hedges' g	0.43		small

The CI of d inverts the noncentral t distribution (Steiger & Fouladi, 1997).

Figure 18. Group descriptives, the t-tests and the effect size.

The result (Figure 18) reports Welch's t-test first, because it is accurate whether or not the variances are equal (Delacre et al., 2017), followed by Student's t-test, the effect size (Cohen's d with its confidence interval, and Hedges' g), the Mann-Whitney U test as the rank-based alternative and Levene's and the Brown-Forsythe tests of equal variances.

What it means for our example

Women ($M = 56.57$, $SD = 9.64$, $n = 239$) scored higher than men ($M = 52.38$, $SD = 9.57$, $n = 148$) on Exam_score. The difference of 4.19 points is statistically significant, Welch's $t(313.32) = 4.17$, $p < .001$, 95% CI [2.21, 6.16], and small by Cohen's benchmarks, $d = 0.44$, 95% CI [0.23, 0.64] (Hedges' $g = 0.43$). SaNaML states the size as $|d| = 0.44$ and explains that the sign of d follows the order of the groups: here $d = 0.44$ means that female scored higher than male. Levene's test does not suggest unequal variances, $F(1, 385) = 0.04$, $p = .834$, so Student's t gives the same result, $t(385) = 4.17$. The Mann-Whitney test agrees, $U = 21966.5$, $z = 4.00$, $p < .001$, rank-biserial $r = .24$. Both groups have more than 30 cases, so the t-test is robust to the slight non-normality.

How to report it

"An independent-samples Welch's t-test showed that Exam_score differed between female ($M = 56.57$, $SD = 9.64$) and male ($M = 52.38$, $SD = 9.57$), $t(313.32) = 4.17$, $p < .001$, $d = 0.44$, 95% CI [0.23, 0.64]."

One-tailed tests. Choose **Right-tailed (one-tailed) test (Only "first is greater")** if you predicted before seeing the data that the first group (female) would be higher. The p-value then refers to that one direction only and the confidence interval becomes one-sided (here $[2.53, \infty]$), while t and d are unchanged. SaNaML's sentence now says so: "An independent-samples Welch's t-test (one-tailed: female greater than male) showed that Exam_score was higher for female ($M = 56.57$, $SD = 9.64$) than for male ($M = 52.38$, $SD = 9.57$), $t(313.32) = 4.17$, $p < .001$, $d = 0.44$, 95% CI [0.23, 0.64]." When the result is not significant it says "was not significantly higher". The confidence interval of d stays two-sided. Never switch to a one-tailed test after seeing the direction of the result.

5.2 Paired-samples t-test: Pre_score and Post_score

In a paired design every student has two scores, so the analysis is of the *differences*.

1. Choose **Paired-samples t-test (The same people measured twice)**.
2. In **First measurement**, choose **Pre_score**; in **Second measurement**, choose **Post_score**. The change is second minus first.

Descriptives (complete pairs)					
Measure	n	Mean	SD	Median	
Pre_score	400	59.37	11.65	59.05	
Post_score	400	66.09	13.47	65.70	
Difference (Post_score – Pre_score)	400	6.72	6.37	6.85	

Paired t-test (two-sided)						
t	df	p	Mean difference	95% CI	d_z	CI of d_z
21.12	399	< .001	6.72	[6.10, 7.35]	1.06	[0.93, 1.18]

Rank-based alternative					
Test	V	z	p	Matched rank-biserial r	Method
Wilcoxon signed-rank	73510.5	14.87	< .001	.86	normal approximation

3 pairs with no change were left out of the signed-rank test.

Figure 19. The descriptives of the complete pairs, the paired t-test and the Wilcoxon signed-rank test.

What it means for our example

Post_score (M = 66.09, SD = 13.47) is higher than Pre_score (M = 59.37, SD = 11.65) by 6.72 points on average (SD of the differences 6.37, n = 400 complete pairs). The change is significant and large, $t(399) = 21.12$, $p < .001$, 95% CI of the difference [6.10, 7.35], $d_z = 1.06$, 95% CI [0.93, 1.18]. The Wilcoxon signed-rank test agrees, $V = 73510.5$, $z = 14.87$, $p < .001$, matched rank-biserial $r = .86$; three students with no change were left out of it.

Effect size d_z is the mean change divided by the SD of the changes (here $6.72 / 6.37 \approx 1.06$). It is not comparable with Cohen's d from independent groups, because its denominator is smaller when the two measurements are strongly correlated (here $r = .88$). The same benchmarks (.2, .5, .8) are applied.

How to report it

"A paired-samples t-test showed that Post_score (M = 66.09, SD = 13.47) differed from Pre_score (M = 59.37, SD = 11.65), $t(399) = 21.12$, $p < .001$, $d_z = 1.06$, 95% CI [0.93, 1.18]."

Note: a rise from pre to post is not proof of a teaching effect

Without a comparison group, a pre-post rise can reflect practice, maturation or the test itself. The comparison of teaching methods in chapter 6 is the stronger design because students were assigned at random.

5.3 One-sample t-test: Exam_score against the pass mark

Passing needs an Exam_score of 50 or more. Is the average above the pass mark?

1. Choose **One-sample t-test (One score against a fixed value)**.
2. In **Score to compare**, choose **Exam_score**, and set **Test value** to 50.

What it means for our example

The mean of Exam_score (54.97, SD = 9.81, n = 387) is significantly above 50: $t(386) = 9.96$, $p < .001$, 95% CI of the mean [53.99, 55.95], $d = 0.51$, a medium effect (95% CI [0.40, 0.61]). The Wilcoxon signed-rank test agrees, $V = 56396.0$, $z = 8.81$, $p < .001$, matched rank-biserial $r = .52$.

How to report it

"A one-sample t-test showed that Exam_score ($M = 54.97$, $SD = 9.81$) differed from the test value of 50.00, $t(386) = 9.96$, $p < .001$, $d = 0.51$."

Tip: choose a meaningful test value

The test value should come from your question, not from your data: a pass mark, the scale midpoint (4 on a 1 to 7 scale), a published norm. Testing a mean against a value just below it will always find a "significant" result with a large sample, so also look at the confidence interval and d .

Exercise: t-tests on practice_marketing.csv and practice_hr.csv

(a) In practice_marketing.csv (1,500 customers), run a paired-samples t-test of Monthly_spend_before and Monthly_spend_after. Answer: spend rose from $M = 282.64$ ($SD = 198.64$) to $M = 304.35$ ($SD = 215.75$), $t(1499) = 28.78$, $p < .001$, $d_z = 0.74$, a medium effect. (b) Compare Annual_spend between loyalty members and non-members (Loyalty_member). Annual_spend is right-skewed and Levene's test is significant, so read Welch's test and the Mann-Whitney test: members spent more, $M = 4389.38$ ($SD = 2591.22$) against $M = 2637.02$ ($SD = 1893.02$), Welch's $t(1131.47) = -14.51$, $p < .001$, $d = -0.79$ (the sign shows the order of the groups), Mann-Whitney $z = -14.09$, $p < .001$. (c) In practice_hr.csv (1,200 employees), run a paired-samples t-test of Skills_test_before and Skills_test_after: $M = 58.13$ ($SD = 11.07$) rose to $M = 63.74$ ($SD = 12.74$), $t(1199) = 31.37$, $p < .001$, $d_z = 0.91$.

6. Comparing three or more groups

This chapter compares the mean of one outcome across three or more independent groups. Students in `practice_students.csv` were assigned at random to one of three teaching methods (Lecture, Blended, Flipped), so a difference between the methods can be read as an effect of teaching, apart from chance.

Concept: why not several t-tests?

With three groups there are three pairs. Testing each pair at $\alpha = .05$ raises the chance of at least one false positive to about 14%, and with five groups (ten pairs) to about 40%. A one-way **analysis of variance (ANOVA)** first asks one question: do the group means differ at all? It compares the variation *between* the group means with the variation *within* the groups (the ratio F). Only if that omnibus test is significant do you look at which pairs differ, with a method that keeps the overall error rate at 5% (post hoc tests).

Note: the post-test and the gain

The planted effect in `practice_students.csv` is a larger *gain* (Post_score minus Pre_score) with the flipped classroom. This chapter compares Post_score across the methods; Section 6.5 analyses the gain itself, computed in Section 1.6, and Section 8.2 uses the pre-test as a covariate, which answers the same question and is usually more powerful.

6.1 One-way ANOVA: Post_score by Teaching_method

1. Choose **One-way ANOVA (Compare three or more groups)** in the Analyses menu.
2. In **Score to compare**, choose **Post_score**; in **Groups**, choose **Teaching_method**.

The result gives the group descriptives, the ANOVA table, Welch's F with η^2 and ω^2 , pairwise comparisons (Tukey-Kramer, or Games-Howell when Levene's test shows unequal variances), the Kruskal-Wallis test with Dunn's pairwise comparisons (Holm-adjusted) as the rank-based alternative (Figure 20).

Group descriptives					
Group	n	Mean	SD		
Blended	129	68.34	11.97		
Flipped	129	68.49	12.53		
Lecture	142	61.87	14.59		

One-way ANOVA					
Source	SS	df	MS	F	p
Between groups	3926.91	2	1963.46	11.38	< .001
Within groups	68473.40	397	172.48		
Total	72400.32	399			

Robust test and effect sizes	
Measure	Value
Welch's F	$F(2, 264.52) = 10.27, p < .001$
η^2	.054
ω^2 (less biased)	.049
Levene's test	$F(2, 397) = 1.76, p = .173$

Figure 20. Group descriptives, the ANOVA table and the robust test with effect sizes.

What it means for our example

Post_score differs across the three groups: Blended $M = 68.34$ ($SD = 11.97, n = 129$), Flipped $M = 68.49$ ($SD = 12.53, n = 129$) and Lecture $M = 61.87$ ($SD = 14.59, n = 142$). $F(2, 397) = 11.38, p < .001, \eta^2 = .054, \omega^2 = .049$, a small effect by the benchmarks. Levene's test does not suggest unequal variances, $F(2, 397) = 1.76, p = .173$, so the classic ANOVA and the Tukey-Kramer comparisons are used; Welch's F agrees, $F(2, 264.52) = 10.27, p < .001$.

6.2 Which groups differ? Post hoc tests

Pairwise comparisons: Tukey-Kramer			
Pair	Mean difference	95% CI	p (adjusted)
Blended – Flipped	−0.15	[−4.00, 3.70]	.995
Blended – Lecture	6.47	[2.71, 10.23]	< .001
Flipped – Lecture	6.62	[2.86, 10.38]	< .001

Figure 21. The Tukey-Kramer pairwise comparisons.

The Tukey–Kramer comparisons (Figure 21) give the difference between each pair of means with a confidence interval and a p-value adjusted for the three comparisons. Blended and Flipped do not differ (difference -0.15 , 95% CI $[-4.00, 3.70]$, $p = .995$); both are higher than Lecture (Blended – Lecture 6.47 , 95% CI $[2.71, 10.23]$, $p < .001$; Flipped – Lecture 6.62 , 95% CI $[2.86, 10.38]$, $p < .001$). Read the confidence intervals: a difference of 6 to 7 points is the best estimate, but the data are compatible with anything from about 3 to 10.

6.3 The rank-based alternative: Kruskal–Wallis and Dunn

The Kruskal–Wallis test compares the groups' rank distributions: $H = 18.32$, $df = 2$, $p < .001$, $\epsilon^2 = .046$. Dunn's pairwise comparisons, adjusted by Holm's method, give the same picture: Blended and Flipped do not differ ($p = .953$), and both differ from Lecture ($p < .001$). When the ANOVA and Kruskal–Wallis tests agree, as here, your conclusion does not depend on the assumptions.

How to report it

"A one-way ANOVA showed a significant effect of Teaching_method on Post_score, $F(2, 397) = 11.38$, $p < .001$, $\eta^2 = .054$, $\omega^2 = .049$. Tukey–Kramer comparisons showed that Blended ($M = 68.34$) and Flipped ($M = 68.49$) students scored higher than Lecture students ($M = 61.87$), both $p < .001$, whereas Blended and Flipped did not differ, $p = .995$."

6.4 Unequal variances: Welch's ANOVA and Games–Howell

Variances were equal in practice_students.csv. To see what SaNaML does when they are not, open practice_hr.csv (1,200 simulated employees) and compare Annual_salary across Department. Levene's test is significant, $F(4, 1195) = 5.10$, $p < .001$, so SaNaML switches to Welch's ANOVA and the Games–Howell comparisons (Figure 22) and says so in the check *Equal variances*.

Pairwise comparisons: Games–Howell (variances differ)			
Pair	Mean difference	95% CI	p (adjusted)
Finance – HR	3172.99	$[-1453.77, 7799.74]$	.329
Finance – IT	–4565.05	$[-9160.06, 29.96]$	.052
Finance – Operations	6616.43	$[3077.50, 10155.37]$	$< .001$
Finance – Sales	3839.12	$[220.50, 7457.75]$	.031
HR – IT	–7738.04	$[-12633.35, -2842.72]$	$< .001$
HR – Operations	3443.45	$[-486.21, 7373.10]$	.116
HR – Sales	666.14	$[-3334.72, 4666.99]$	.991
IT – Operations	11181.48	$[7294.28, 15068.69]$	$< .001$
IT – Sales	8404.17	$[4444.37, 12363.98]$	$< .001$
Operations – Sales	–2777.31	$[-5424.41, -130.21]$	.034

Figure 22. Games–Howell pairwise comparisons of Annual_salary across five departments (practice_hr.csv).

What it means for our example

Welch's $F(4, 439.57) = 18.43$, $p < .001$, $\omega^2 = .066$ (a medium effect). IT employees earn the most ($M = 52486.00$, $SD = 17533.10$, $n = 195$) and Operations the least ($M = 41304.52$, $SD = 12491.95$, $n = 363$). The largest difference is IT – Operations = 11181.48, 95% CI [7294.28, 15068.69], Games-Howell $p < .001$. The Kruskal-Wallis test agrees, $H = 87.86$, $df = 4$, $p < .001$.

Games-Howell is the post hoc test of choice when variances or group sizes differ, because it does not pool the variances (Games & Howell, 1976). When the variances are equal, Tukey-Kramer is slightly more powerful.

6.5 The gain score

The variable Gain (Post_score – Pre_score, Section 1.6) is the outcome that the planted effect was built on. Choose **One-way ANOVA**, **Gain** as **Score to compare** and **Teaching_method** as **Groups** (Figure 23). The mean gain is 7.51 points for Blended ($SD = 6.03$, $n = 129$), 8.93 for Flipped ($SD = 6.11$, $n = 129$) and 4.01 for Lecture ($SD = 5.94$, $n = 142$); $F(2, 397) = 24.15$, $p < .001$, $\eta^2 = .108$, $\omega^2 = .104$, a medium effect. Levene's test does not suggest unequal variances, $F(2, 397) = 0.08$, $p = .922$. The Tukey-Kramer comparisons show that Flipped gains 4.92 points more than Lecture, 95% CI [3.20, 6.64], and Blended 3.50 points more, 95% CI [1.78, 5.23] (both $p < .001$), while Blended and Flipped do not differ, -1.42 , 95% CI $[-3.18, 0.35]$, $p = .142$. The effect is larger than for Post_score alone ($\omega^2 = .049$): taking the pre-test into account removes the differences between students that were there before the course.

Group descriptives				
Group	n	Mean	SD	
Blended	129	7.51	6.03	
Flipped	129	8.93	6.11	
Lecture	142	4.01	5.94	

One-way ANOVA					
Source	SS	df	MS	F	p
Between groups	1753.60	2	876.80	24.15	< .001
Within groups	14416.26	397	36.31		
Total	16169.86	399			

Robust test and effect sizes	
Measure	Value
Welch's F	$F(2, 263.04) = 24.24, p < .001$
η^2	.108
ω^2 (less biased)	.104
Levene's test	$F(2, 397) = 0.08, p = .922$

Figure 23. The one-way ANOVA of Gain by Teaching_method.

How to report it

"A one-way ANOVA showed a significant effect of Teaching_method on Gain, $F(2, 397) = 24.15, p < .001, \eta^2 = .108, \omega^2 = .104$."

Exercise: ANOVA on three practice files

(a) practice_marketing.csv: does Satisfaction_1to10 differ by Main_channel? Answer: no, $F(2, 1412) = 1.55, p = .213, \omega^2 = .001$. (b) practice_hr.csv: does Engagement differ by Department? Answer: no, $F(4, 1195) = 0.28, p = .891$. (c) practice_engagement.csv: build the scale Dedication (Section 1.5, exercise) and compare it across Department. Answer: it differs, $F(4, 595) = 10.84, p < .001, \eta^2 = .068, \omega^2 = .062$; HR has the highest mean (4.67) and Operations the lowest (3.57). Which pairs differ in the Tukey-Kramer table? When a result is not significant, what does the confidence interval of the difference tell you?

7. The same people measured repeatedly

In `practice_students.csv` each student sat the same test three times: `Score_T1`, `Score_T2` and `Score_T3`. The scores of one student are correlated, so they are not independent groups. The one-way repeated-measures ANOVA compares the three means while using each student as their own control.

Concept: sphericity

Repeated-measures ANOVA assumes **sphericity**: the variances of the differences between every pair of measurements are equal. If it fails, the F test is too liberal. **Mauchly's test** (Mauchly, 1940) checks it, and the **epsilon (ϵ)** of Greenhouse–Geisser (1959) or Huynh–Feldt (1976) measures how far it fails (1 = perfect; lower is worse). The corrections multiply the degrees of freedom by ϵ . With only two measurements sphericity is always met.

7.1 Run the analysis

1. Choose **One-way repeated-measures ANOVA (Same people, three or more times)**.
2. Tick **Score_T1**, **Score_T2** and **Score_T3** under *Measurements*. SaNaML orders them as they appear in the file, which should be the order of measurement. Respondents with any missing measurement are left out (here nobody: $n = 400$).

The result gives the descriptives of the complete cases, the repeated-measures ANOVA with the correction, Mauchly's test, pairwise paired t-tests (Holm-adjusted) and the Friedman test with Wilcoxon pairwise tests.

Descriptives (complete cases)				
Measure	n	Mean	SD	
Score_T1	400	59.37	11.65	
Score_T2	400	62.55	12.66	
Score_T3	400	66.09	13.47	

Repeated-measures ANOVA				
Correction	df	F	p	ϵ
None (sphericity assumed)	2, 798	346.88	< .001	1
Greenhouse-Geisser	1.53, 610.27	346.88	< .001	.765
Huynh-Feldt	1.53, 612.14	346.88	< .001	.767

Partial $\eta^2 = .465$; generalised $\eta^2 = .045$ (Bakeman, 2005).

Mauchly's test of sphericity				
W	χ^2	df	p	
0.692	146.32	2	< .001	

Figure 24. The descriptives, the repeated-measures ANOVA and Mauchly's test.

What it means for our example

The mean score rises from 59.37 (SD = 11.65) at T1 through 62.55 (SD = 12.66) at T2 to 66.09 (SD = 13.47) at T3. Mauchly's test is significant, $W = 0.692$, $\chi^2(2) = 146.32$, $p < .001$, and ϵ is .765 (Greenhouse-Geisser) and .767 (Huynh-Feldt). Because $\epsilon \geq .75$, SaNaML uses the Huynh-Feldt correction: $F(1.53, 612.14) = 346.88$, $p < .001$, partial $\eta^2 = .465$, generalised $\eta^2 = .045$.

Note: partial and generalised η^2

Partial $\eta^2 = .465$ is large, but SaNaML labels the effect "small" because it applies the benchmarks to the generalised η^2 of .045. The generalised η^2 keeps the differences between students in its denominator and is therefore always much smaller than the partial η^2 in a within-subject design. It is meant for comparing designs (Bakeman, 2005); report both, and judge the size of the change from the pairwise effect sizes below.

7.2 Where is the change? Pairwise comparisons

Each pair of time points differs: T1 – T2 = -3.18, $t(399) = -14.94$, $d_z = -0.75$; T1 – T3 = -6.72, $t(399) = -21.12$, $d_z = -1.06$; T2 – T3 = -3.54, $t(399) = -16.00$, $d_z = -0.80$; all Holm-adjusted $p < .001$. The negative sign only shows that the earlier measurement is lower. The scores rise steadily, by about three to four points between occasions.

7.3 The rank-based alternative: Friedman

The Friedman test ranks each student's three scores and compares the mean ranks: $\chi^2(2) = 327.05$, $p < .001$, Kendall's $W = .409$, with all three Wilcoxon pairwise tests significant after Holm's adjustment (r between $-.72$ and $-.86$). It leads to the same conclusion, so the result does not depend on the sphericity or normality assumptions.

How to report it

"A one-way repeated-measures ANOVA with the Huynh–Feldt correction ($\epsilon = .767$) showed a significant difference across the three measurements, $F(1.53, 612.14) = 346.88$, $p < .001$, partial $\eta^2 = .465$, generalised $\eta^2 = .045$. Holm-adjusted paired t-tests showed that every measurement was higher than the one before (all $p < .001$)."

Note: what this screen cannot do

SaNaML offers the one-way repeated-measures ANOVA only. A design that combines repeated measures with a between-group factor (a mixed ANOVA, for example the three scores by teaching method) is not available; for such designs use a different tool, or compare the groups at each occasion with the tests of chapter 6.

Exercise: repeated measures in practice_hr.csv

Open practice_hr.csv and tick Engagement_Q1 to Engagement_Q4. Answer: the quarters differ, $F(3, 3597) = 15.79$, $p < .001$, partial $\eta^2 = .013$, generalised $\eta^2 = .002$, which SaNaML calls negligible. With 1,200 employees even a tiny change is significant, so read the effect size, not the p-value, to judge whether the change matters. Which pairs of quarters differ?

8. Factorial ANOVA, ANCOVA and MANOVA

Real outcomes usually depend on more than one thing. This chapter covers three extensions of the ANOVA: two or more grouping factors (factorial ANOVA), a covariate (ANCOVA), and two or more outcomes (MANOVA). All three are under **Factorial ANOVA, ANCOVA and MANOVA (Several factors, covariates or outcomes)**.

Concept: main effects and interaction

A **main effect** is the effect of one factor averaged over the levels of the other. An **interaction** means that the effect of one factor depends on the level of the other (for example, the flipped classroom helps women but not men). If there is an interaction, interpret the cell means, not the main effects alone. SaNaML reports Type III sums of squares with sum-to-zero coding, so each effect is tested after all the others, which suits groups of unequal size, and gives partial η^2 (the share of the variance not explained by other effects that the effect explains).

8.1 Factorial ANOVA: Exam_score by Teaching_method and Gender

1. Under *What is your design?* choose **Factorial ANOVA / ANCOVA**.
2. As **Score (outcome)** choose **Exam_score**, and tick **Teaching_method** and **Gender** under **Grouping factors**.
3. Keep **Full factorial model (Include the interaction; usual)**. **Main-effects model** leaves the interaction out and is for very small cells.

Cell descriptives			
Cell	n	Mean	SD
female · Blended	78	59.01	9.41
female · Flipped	72	56.34	8.32
female · Lecture	89	54.61	10.43
male · Blended	46	52.85	9.51
male · Flipped	50	53.84	9.91
male · Lecture	52	50.56	9.17

Factorial ANOVA (Type III sums of squares)						
Source	SS	df	MS	F	p	Partial η^2
Gender	1633.48	1	1633.48	18.06	< .001	.045
Teaching_method	757.63	2	378.81	4.19	.016	.022
Gender × Teaching_method	195.86	2	97.93	1.08	.340	.006
Error	34460.78	381	90.45			
Total (corrected)	37159.76	386				

$R^2 = .073$, adjusted $R^2 = .060$, $n = 387$.

Figure 25. The cell descriptives and the ANOVA table with partial η^2 .

The result (Figure 25) checks that the variances are equal across the six cells (Levene's test on the medians) and that the cells are large enough, and then gives the cell descriptives, the ANOVA table, and for each factor the estimated marginal means and Holm-adjusted pairwise comparisons.

What it means for our example

Levene's test does not suggest unequal variances across the six cells, $F(5, 381) = 1.03$, $p = .401$, and the smallest cell has 46 students. There is a significant main effect of Gender, $F(1, 381) = 18.06$, $p < .001$, partial $\eta^2 = .045$, and of Teaching_method, $F(2, 381) = 4.19$, $p = .016$, partial $\eta^2 = .022$, but no interaction, $F(2, 381) = 1.08$, $p = .340$, partial $\eta^2 = .006$: the advantage of women is about the same under all three methods (the cell means are 59.01, 56.34 and 54.61 for women and 52.85, 53.84 and 50.56 for men under Blended, Flipped and Lecture). The model explains 7.3% of the variance ($R^2 = .073$, adjusted $R^2 = .060$, $n = 387$ complete cases).

The **estimated marginal means** are the group means after allowing for the other factor: 56.65 for women (95% CI [55.44, 57.87]) and 52.42 for men (95% CI [50.88, 53.96]), a difference of 4.24 (SE = 0.997), $t(381) = 4.25$, $p < .001$ (Holm). For teaching methods the marginal means are Blended 55.93, Flipped 55.09 and Lecture 52.59; only Blended – Lecture differs (3.34, $p = .018$), while Blended – Flipped ($p = .502$) and Flipped – Lecture ($p = .077$) do not.

How to report it

"A 2 (Gender) $\times$ 3 (Teaching method) factorial ANOVA on Exam_score showed a significant effect of Gender, $F(1, 381) = 18.06$, $p < .001$, partial $\eta^2 = .045$, and of Teaching_method, $F(2, 381) = 4.19$, $p = .016$, partial $\eta^2 = .022$; the interaction was not significant, $F(2, 381) = 1.08$, $p = .340$, partial $\eta^2 = .006$."

8.2 ANCOVA: adding the pre-test as a covariate

Tick **Pre_score** under **Covariates (optional: makes it an ANCOVA)** and the title changes to ANCOVA.

What it means for our example

Pre_score is a strong covariate, $F(1, 380) = 201.33$, $p < .001$, partial $\eta^2 = .346$, and the error sum of squares falls from 34460.78 to 22526.27 (R^2 rises from .073 to .394). Gender becomes more important, $F(1, 380) = 31.64$, $p < .001$, partial $\eta^2 = .077$; the adjusted means are 56.74 for women and 52.20 for men (difference 4.54, SE = 0.807, $t(380) = 5.63$, $p < .001$). Teaching_method is no longer significant, $F(2, 380) = 2.62$, $p = .074$, partial $\eta^2 = .014$, and neither is the interaction, $F(2, 380) = 2.82$, $p = .061$, partial $\eta^2 = .015$.

The exam score is a single test, taken after the course, and the pre-test explains most of why some students score higher; after that, the teaching method adds only a little (the comparison of chapter 6, which used the post-test, found a clearer effect). Covariates change what a factor's p-value answers: it is now the effect of the method *for students with the same pre-test score*. Do not choose between a factorial ANOVA and an ANCOVA by which one gives the p-value you prefer: decide the model before you look.

Cut-off values: ANCOVA assumptions

The covariate must be measured *before* the groups were treated (the pre-test is), it should be related to the outcome, and the relationship must have the same slope in every group (no covariate $\times$ group interaction). SaNaML does not test equal slopes: check it with the scatter plot of Post_score against Pre_score grouped by Teaching_method (Section 3.6) and look for roughly parallel lines (Maxwell et al., 2018). SaNaML's ANCOVA sentence begins "A ANCOVA"; write "An ANCOVA" in your report.

8.3 One-way MANOVA: Exam_score and Project_score together

Students have two outcomes, Exam_score and Project_score. A **multivariate analysis of variance (MANOVA)** tests whether the groups differ on the outcomes *taken together*, which controls the error rate across the outcomes and uses their correlation.

1. Under *What is your design?* choose **One-way MANOVA (Several scores at once, one grouping factor)**.
2. Tick **Exam_score** and **Project_score** under **Outcomes**, and choose **Teaching_method** as **Grouping factor**.

Multivariate tests					
Test	Value	F	df1	df2	p
Pillai's trace	0.126	12.91	4	768	< .001
Wilks' lambda	0.876	13.09	4	766	< .001
Hotelling-Lawley trace	0.139	13.27	4	764	< .001
Roy's largest root	0.118	22.74	2	384	< .001
n = 387 complete cases. Roy's largest root gives an upper bound for F.					
Follow-up: each outcome separately					
Outcome	F	df	p	Partial η^2	
Exam_score	4.65	2, 384	.010	.024	
Project_score	21.03	2, 384	< .001	.099	
Consider a Bonferroni-adjusted α (α divided by the number of outcomes) for these follow-up tests.					

Figure 26. The multivariate tests and the follow-up ANOVAs.

What it means for our example

The three teaching methods differ on the two outcomes taken together: Pillai's trace $V = .126$, $F(4, 768) = 12.91$, $p < .001$ ($n = 387$ complete cases; Wilks' lambda = .876, $F(4, 766) = 13.09$, $p < .001$; the other two tests agree). The follow-up ANOVAs (Figure 26) show an effect on both outcomes, but a much larger one on Project_score, $F(2, 384) = 21.03$, $p < .001$, partial $\eta^2 = .099$, than on Exam_score, $F(2, 384) = 4.65$, $p = .010$, partial $\eta^2 = .024$.

Cut-off values: MANOVA

Report Pillai's trace: it is the most robust of the four multivariate tests when groups are unequal or the assumptions are doubtful (Olson, 1976). SaNaML suggests a Bonferroni-adjusted α for the follow-up tests, the α divided by the number of outcomes: here $.05 / 2 = .025$, and both follow-up p-values (.010 and $< .001$) are below it. SaNaML does not test the equality of the covariance matrices; with groups of similar size (124, 122 and 141 complete cases) the MANOVA is reasonably robust.

How to report it

"A one-way MANOVA showed a significant multivariate effect of Teaching method on Exam_score and Project_score, Pillai's $V = .126$, $F(4, 768) = 12.91$, $p < .001$. Follow-up ANOVAs ($\alpha = .025$) showed significant effects on Exam_score, $F(2, 384) = 4.65$, $p = .010$, partial $\eta^2 = .024$, and on Project_score, $F(2, 384) = 21.03$, $p < .001$, partial $\eta^2 = .099$."

Exercise: factorial designs

In practice_students.csv, run a factorial ANOVA of Project_score by Teaching_method and Gender. Is there an interaction? Then add Pre_score as a covariate: does the effect of Teaching_method change? Compare the cell means with the main effects before you decide how to describe the result. In practice_hr.csv you can run a factorial ANOVA of Annual_salary by Gender and Department; the file note says that women earn about 4% less after controls, so look at Gender with Department in the model.

9. Categories and proportions

This chapter covers variables that are counts of categories rather than scores: whether two categorical variables are related (the chi-square test of independence), whether the categories of one variable are equally common (the goodness-of-fit test) and whether a yes/no answer changed in the same people (the McNemar test).

Concept: expected counts and the chi-square statistic

A chi-square test compares the counts you observed in a table with the counts you would *expect* if the variables were unrelated. The expected count of a cell is its row total times its column total divided by N . The larger the gap between observed and expected, the larger χ^2 , and the smaller the p-value. As with every test, p says nothing about the size of the association: report an effect size, Cramér's V (ϕ for a 2×2 table), with it.

Choose **Chi-square, Fisher's exact and McNemar tests (Categories and proportions)** in the Analyses menu. The question **What do you want to know?** offers the three tests of this chapter.

9.1 Chi-square test of independence: Gender and Programme

1. Choose **Chi-square test of independence (Are two categorical variables related?)**.
2. In **Categorical variable** (the rows), choose **Gender**; in **Second categorical variable** (the columns), choose **Programme** (Figure 27).

The screenshot shows the 'What do you want to know?' dialog box for the 'Chi-square, Fisher's exact and McNemar tests (Categories and proportions)' menu. It contains three radio button options: 'Chi-square test of independence (Are two categorical variables related?)' (selected), 'Chi-square goodness-of-fit test (Are the categories of one variable equally common?)', and 'McNemar test (Did the same yes/no answer change over time?)'. Below these is a dropdown menu for 'Categorical variable' with 'Gender (2 groups)' selected. At the bottom, there is a section for 'Second categorical variable'.

Figure 27. The chi-square test of independence: Gender by Programme.

The check on the expected counts comes first: no expected count should be below 1 and no more than 20% should be below 5 (Cochran, 1954). Here the smallest expected count is 35.42, so the test is trustworthy. The tables (Figure 28) show the counts with row percentages, Pearson's chi-square and the likelihood-ratio test, and the effect size.

Counts (row %)				
Gender	Arts	Business	Science	Total
female	66 (27%)	91 (37%)	89 (36%)	246
male	26 (17%)	72 (47%)	56 (36%)	154

Tests			
Test	Value	df	p
Pearson chi-square	6.29	2	.043
Likelihood ratio	6.43	2	.040

Figure 28. The counts with row percentages, and the tests.

Row percentages answer the question "what share of each row falls in each column?". Of the 246 women, 27% study Arts, against 17% of the 154 men. The adjusted standardised residuals (Figure 29) show where the pattern comes from: a value beyond ± 1.96 marks a cell with clearly more (positive) or fewer (negative) cases than expected. Women are over-represented in Arts (2.30) and men under-represented (-2.30).

Adjusted standardised residuals			
Gender	Arts	Business	Science
female	2.30 *	-1.93	-0.04
male	-2.30 *	1.93	0.04

* beyond ± 1.96 : this cell holds clearly more (positive) or fewer (negative) cases than expected.

Effect size		
Measure	Value	Size
Cramér's V	.125	small

Figure 29. The adjusted standardised residuals and the effect size.

What it means for our example

Gender and Programme are related, $\chi^2(2, N = 400) = 6.29$, $p = .043$ (likelihood ratio 6.43, $p = .040$), but weakly: Cramér's $V = .125$, a small effect. The relationship is carried by the Arts programme, which has proportionally more women than the other two. Business comes close to the cut-off (women -1.93, men 1.93) and Science does not differ from the expected counts at all (-0.04 and 0.04).

How to report it

"A chi-square test of independence showed a significant association between Gender and Programme, $\chi^2(2, N = 400) = 6.29$, $p = .043$, Cramér's $V = .13$."

9.2 Teaching_method and Passed: reading row percentages as rates

Change **Categorical variable** to **Teaching_method** and **Second categorical variable** to **Passed** (Figure 30). Now the row percentages are pass rates: 78% of the 129 students taught in the blended format passed, 71% of the 129 in the flipped format and 60% of the 142 in the lecture format. The association is significant, $\chi^2(2, N = 400) = 11.13$, $p = .004$, with Cramér's $V = .167$ (small). The residuals show that the lecture group has more students who failed than expected (3.11) and the blended group fewer (-2.64); the flipped group is close to the expected counts (-0.54).

Counts (row %)			
Teaching_method	No	Yes	Total
Blended	28 (22%)	101 (78%)	129
Flipped	37 (29%)	92 (71%)	129
Lecture	57 (40%)	85 (60%)	142

Tests			
Test	Value	df	p
Pearson chi-square	11.13	2	.004
Likelihood ratio	11.15	2	.004

Adjusted standardised residuals		
Teaching_method	No	Yes
Blended	-2.64 *	2.64 *
Flipped	-0.54	0.54
Lecture	3.11 *	-3.11 *

* beyond ± 1.96 : this cell holds clearly more (positive) or fewer (negative) cases than expected.

Figure 30. Counts and residuals for Teaching_method by Passed.

Put the grouping variable in the rows and the outcome in the columns, so that the row percentages are the rates you want to compare.

How to report it

"A chi-square test of independence showed a significant association between Teaching_method and Passed, $\chi^2(2, N = 400) = 11.13$, $p = .004$, Cramér's $V = .17$. The pass rate was 78% for the blended, 71% for the flipped and 60% for the lecture format."

9.3 Two-by-two tables: Gender and Passed

With two rows and two columns SaNaML adds the continuity-corrected (Yates) chi-square, Fisher's exact test and ϕ . Of the 246 women, 76% passed, against 58% of the 154 men. Pearson's chi-square is $\chi^2(1, N = 400) = 14.45$, $p < .001$; the Yates correction gives 13.61 and Fisher's exact test is also below .001; Cramér's $V = .190$. All versions agree, as they do whenever the expected counts are large (the smallest is 46.97). Fisher's exact test matters when the expected counts are small, for example below 5 in one cell.

9.4 Goodness of fit: are the programmes equally common?

1. Choose **Chi-square goodness-of-fit test (Are the categories of one variable equally common?)**.
2. In **Categorical variable**, choose **Programme** (Figure 31).



Figure 31. The goodness-of-fit test for Programme.

The test compares the observed counts (Arts 92, Business 163, Science 145) with equal shares, 133.3 each. The categories are not equally common, $\chi^2(2, N = 400) = 20.44$, $p < .001$, with Cohen's $w = .23$ (a small to medium effect by Cohen's benchmarks of .10, .30 and .50). SaNaML tests against equal proportions only; to test against other known proportions, use the observed and expected counts to compute the statistic by hand.

How to report it

"A chi-square goodness-of-fit test against equal proportions was significant, $\chi^2(2, N = 400) = 20.44$, $p < .001$, $w = .23$."

9.5 McNemar test: did the answer change?

In `practice_students.csv`, `Pass_before` and `Pass_after` are the pass/fail versions (pass mark 50) of `Pre_score` and `Post_score`, so they describe the same 400 students before and after the course. A chi-square test of independence would be wrong here because the two answers come from the same people.

1. Choose **McNemar test (Did the same yes/no answer change over time?)**.
2. In **First time (yes/no)**, choose **Pass_before**; in **Second time (yes/no)**, choose **Pass_after**.

McNemar test (Change from Pass_before to Pass_after)

What it means: The share answering "Yes" went from 80% (Pass_before) to 90% (Pass_after): a significant change.

Info Which p-value. Enough respondents changed their answer for the chi-square version.

Paired counts			
Pass_before \ Pass_after	No	Yes	
No	38	43	
Yes	4	315	

Tests		
Test	Value	p
McNemar (continuity-corrected)	30.72	< .001
Exact binomial		< .001

Figure 32. The finding, the paired counts and the tests.

Only the students who changed their answer carry information. Of the 400, 43 changed from No to Yes and 4 from Yes to No; 38 said No and 315 said Yes both times. The share answering Yes went from 80% to 90%. Enough students changed for the chi-square version, $\chi^2(1) = 30.72$, $p < .001$ (continuity-corrected); the exact binomial test also gives $p < .001$. SaNaML says which p-value to use: the exact test is the one to report when only a few respondents changed.

How to report it

"A McNemar test showed a significant change in the share answering Yes, from 80% to 90%, $\chi^2(1) = 30.72$, $p < .001$."

Exercise: categories in the other practice files

(a) In practice_marketing.csv, test Campaign_group (rows) against Responded. Answer: 44% of the 738 customers who received the new offer responded, against 26% of the 762 who received the standard offer, $\chi^2(1, N = 1500) = 54.50$, $p < .001$, Cramér's $V = .19$. (b) Test Churned against Loyalty_member. Answer: 46% of the 1,250 customers who did not churn were loyalty members, against 26% of the 250 who did, $\chi^2(1, N = 1500) = 35.64$, $p < .001$, $V = .15$. (c) In practice_hr.csv, test Department against Left_company. Answer: no significant association, $\chi^2(4, N = 1200) = 8.42$, $p = .077$, $V = .08$, although the residual for Operations (2.80) is beyond ± 1.96 : an example of a cell that should not be interpreted when the whole table is not significant.

10. Correlation and regression

This chapter asks how variables go together and which of them predict an outcome. Correlation describes the strength and direction of a relationship; regression predicts an outcome from one or more predictors and says how much each contributes. Both are under **Correlation and regression analysis (How variables go together and predict an outcome)**.

Concept: correlation is not causation

A correlation of $r = .27$ between study hours and exam scores says that students who study more tend to score higher. It does not say that studying more *causes* higher scores: a third variable (ability, motivation) can produce the same pattern, and the direction could be reversed. Correlation and regression describe association; causal claims need a design that supports them (such as the random assignment of teaching methods).

10.1 Pearson correlation

1. Choose **Bivariate and partial correlation (How strongly do variables go together?)**.
2. Tick **Study_hours_week**, **Sleep_hours**, **Anxiety_score** and **Exam_score** under *Variables*, and keep **Pearson product-moment correlation (Pearson r)**.

Correlation matrix					
	1	2	3	4	
1. Study_hours_week	—				
2. Sleep_hours	-.20***	—			
3. Anxiety_score	.12*	-.31***	—		
4. Exam_score	.27***	.20***	-.12*	—	
* p < .05, ** p < .01, *** p < .001 (Holm-adjusted). Each pair uses every respondent with both values.					
Every pair					
Pair	r	95% CI	n	p	p (Holm)
Study_hours_week - Sleep_hours	-.201	[-.29, -.10]	388	< .001	< .001
Study_hours_week - Anxiety_score	.124	[.02, .22]	378	.016	.033
Study_hours_week - Exam_score	.273	[.18, .36]	387	< .001	< .001
Sleep_hours - Anxiety_score	-.306	[-.40, -.21]	366	< .001	< .001
Sleep_hours - Exam_score	.204	[.10, .30]	376	< .001	< .001
Anxiety_score - Exam_score	-.122	[-.22, -.02]	365	.020	.033

Figure 33. The correlation matrix and the table of every pair (Pearson).

The result shows the correlation matrix with stars for Holm-adjusted significance and a table of every pair with its 95% confidence interval, n, p and Holm-adjusted p. Each pair uses every student with both values (so n differs between pairs: 365 to 388).

What it means for our example

All six correlations are significant after Holm's adjustment. The strongest is Sleep_hours with Anxiety_score, $r = -.31$ (medium; students who sleep more are less anxious). Exam_score correlates with Study_hours_week, $r = .27$, 95% CI [.18, .36], $n = 387$, and with Sleep_hours, $r = .20$, 95% CI [.10, .30], and negatively with Anxiety_score, $r = -.12$, 95% CI [-.22, -.02], Holm-adjusted $p = .033$. Study_hours_week and Anxiety_score are barely related, $r = .12$ (Holm-adjusted $p = .033$).

How to report it

"Exam_score correlated positively with Study_hours_week, $r(385) = .27$, $p < .001$, 95% CI [.18, .36], and Sleep_hours, $r(374) = .20$, $p < .001$, 95% CI [.10, .30], and negatively with Anxiety_score, $r(363) = -.12$, Holm-adjusted $p = .033$ (Pearson correlations, Holm-adjusted p -values)."

10.2 Spearman and Kendall: rank correlations

Pearson's r measures a *straight-line* relationship and is pulled by extreme values. Spearman's ρ and Kendall's τ -b use ranks: they describe any steadily rising or falling relationship and are not affected by how extreme the extreme values are. Choose **Spearman's rank-order correlation (Spearman's ρ)** or **Kendall's rank correlation (Kendall's τ -b)**.

What it means for our example

The rank correlation of Study_hours_week with Exam_score is much larger than the Pearson correlation: $\rho = .46$, 95% CI [.37, .54], against $r = .27$ (Kendall's τ -b = .32, 95% CI [.26, .38]). The relationship is steadily rising but not a straight line, and the five extreme study times (Section 3.4) probably have a large influence on Pearson's r . In the other direction, the small Pearson correlation of Study_hours_week with Anxiety_score ($r = .12$, adjusted $p = .033$) is not supported by the ranks ($\rho = .03$, $p = .611$), which suggests that it came from the extreme values. When the two kinds of correlation disagree like this, report both and say why.

Use Spearman's ρ for ordinal data (single Likert items), for skewed variables and when outliers are present. Kendall's τ -b is preferable with many tied values or small samples; its values are smaller than ρ for the same data.

10.3 Partial correlation

A partial correlation is the correlation of two variables after removing what a third variable explains in both. Tick **Pre_score** under **Control for (optional: partial correlation)**; the title becomes *Partial correlation (Pearson r , controlling for Pre_score)*.

Partial correlations use complete cases only ($n = 354$ for every pair here). Controlling for the pre-test, Sleep_hours correlates more strongly with Exam_score ($r = .29$, 95% CI [.19, .39], against .20 before) and Anxiety_score correlates more negatively with it ($r = -.19$, against $-.12$): part of the

noise in Exam_score was prior ability. Study_hours_week with Exam_score remains $r = .27$. Control only for variables that make sense as common causes or confounders, not for variables that lie on the path between the two.

10.4 Multiple linear regression

Concept: regression

Regression fits the equation $outcome = intercept + B_1 \times predictor_1 + B_2 \times predictor_2 + \dots + error$. Each unstandardised coefficient **B** is the change in the outcome for a one-unit change in that predictor, holding the others constant; the standardised β puts the predictors on a common scale (in SD units) so that they can be compared. R^2 is the share of the variance in the outcome that the predictors explain together; adjusted R^2 corrects it for the number of predictors. Effect size for the model: $f^2 = R^2 / (1 - R^2)$.

1. Choose **Multiple linear regression (Which variables predict a score?)**.
2. As **Outcome (scale)** choose **Exam_score**.
3. Tick **Study_hours_week**, **Sleep_hours** and **Anxiety_score** under **Block 1 predictors**.
Categorical variables are marked "(groups)" and are dummy-coded against their first group.

The result checks the sample size, collinearity (VIF), normality of the residuals, constant error variance (Breusch–Pagan), influential cases (Cook's distance) and independent errors (Durbin–Watson), and then gives the model summary and the coefficients (Figure 34). The regression uses only complete cases: $n = 354$ of 400 (46 students with a missing value on any of the four variables are left out).

Model summary							
Model	Predictors	R^2	Adjusted R^2	F	df	p	ΔR^2
1	Study_hours_week, Sleep_hours, Anxiety_score	.142	.134	19.28	3, 350	< .001	

Coefficients (model 1)							
Predictor	B	SE	β	t	p	95% CI of B	VIF
(Intercept)	39.625	4.555		8.70	< .001	[30.668, 48.583]	
Study_hours_week	0.500	0.081	.315	6.20	< .001	[0.342, 0.659]	1.05
Sleep_hours	2.257	0.474	.252	4.76	< .001	[1.324, 3.191]	1.14
Anxiety_score	−0.266	0.171	−.081	−1.55	.121	[−0.602, 0.071]	1.11

n = 354 (46 respondents with missing values left out). Categorical predictors are dummy-coded against their first group. β = standardised coefficient.

Figure 34. The model summary and the coefficients of model 1.

What it means for our example

The three predictors together explain 14.2% of the variance in Exam_score, $R^2 = .142$, adjusted $R^2 = .134$, $F(3, 350) = 19.28$, $p < .001$, $f^2 = 0.17$ (medium). Study_hours_week ($B = 0.500$, $SE = 0.081$, $\beta = .315$, 95% CI of B [0.342, 0.659]) and Sleep_hours ($B = 2.257$, $SE = 0.474$, $\beta = .252$, 95% CI [1.324, 3.191]) are significant predictors: each additional weekly hour of study goes with half a point more, and each additional hour of sleep with 2.3 points more, holding the other predictors constant. Anxiety_score is not significant, $B = -0.266$, $\beta = -.081$, $p = .121$. The largest VIF is 1.14, so collinearity is no problem; the residuals are normal (Shapiro–Wilk $W = 0.997$, $p = .645$); no Cook's distance exceeds 1 (largest 0.773). One check is marked *Note*: the Breusch–Pagan test, $\chi^2(3) = 30.37$, $p < .001$, suggests that the error spread is not constant.

10.5 Hierarchical regression: what does a second block add?

In a hierarchical regression, predictors enter in blocks in an order decided by theory, and the change in R^2 (ΔR^2) tests what the later block adds. Tick **Gender (groups)** under **Block 2 predictors (optional: what do they add?)**. The title becomes *Multiple linear regression (two blocks)* and the model summary has two rows.

What it means for our example

Adding Gender raises R^2 from .142 to .172 (adjusted R^2 .163), $\Delta R^2 = .031$, $p < .001$; the whole model is $F(4, 349) = 18.18$, $p < .001$. Men score 3.48 points lower than women on Exam_score, holding study hours, sleep and anxiety constant: $B = -3.480$, $SE = 0.969$, $\beta = -.176$, $t(349) = -3.59$, $p < .001$, 95% CI $[-5.386, -1.575]$. The coefficients of study hours ($B = 0.481$, $\beta = .302$) and sleep ($B = 2.154$, $\beta = .241$) hardly change, and anxiety remains non-significant ($p = .190$).

How to report it

"A hierarchical multiple linear regression predicted Exam_score from Study_hours_week, Sleep_hours and Anxiety_score (block 1, $R^2 = .142$) and Gender (block 2). Gender added a significant increment, $\Delta R^2 = .031$, $p < .001$, final $F(4, 349) = 18.18$, $p < .001$, $R^2 = .172$, adjusted $R^2 = .163$. Study_hours_week ($\beta = .30$, $p < .001$), Sleep_hours ($\beta = .24$, $p < .001$) and Gender (male vs female, $\beta = -.18$, $p < .001$) were significant predictors."

10.6 Binary logistic regression: predicting Passed

When the outcome has two categories, linear regression is not appropriate; logistic regression models the probability of one of them.

Concept: odds and odds ratios

The **odds** of an event are the probability of the event divided by the probability of its absence (a 75% chance is odds of 3 to 1). A logistic regression coefficient is the change in the *log odds* for a one-unit change in the predictor; its exponential is the **odds ratio (OR)**. $OR = 1$ means no effect; above 1 the odds of the event rise; below 1 they fall. The *event* is the answer coded 1.

1. Choose **Binary logistic regression (Which variables predict a yes/no outcome?)**.
2. As **Outcome (two groups)** choose **Passed**, and choose "Yes" as the event.
3. Tick **Study_hours_week**, **Sleep_hours**, **Anxiety_score** and **Gender (groups)** under *Predictors*.

Model fit						
χ^2	df	p	Nagelkerke R ²	McFadden R ²	Tjur R ²	AUC 95% CI of AUC
63.56	4	< .001	.225	.141	.186	.780 [.73, .83]

Coefficients						
Predictor	B	SE	Wald	p	Odds ratio	95% CI of OR
(Intercept)	-2.740	1.267	4.68	.031	0.065	[0.005, 0.774]
Gender: male (vs female)	-0.906	0.251	13.06	< .001	0.404	[0.247, 0.661]
Study_hours_week	0.254	0.050	25.69	< .001	1.289	[1.168, 1.422]
Sleep_hours	0.485	0.132	13.58	< .001	1.624	[1.255, 2.101]
Anxiety_score	-0.066	0.045	2.19	.139	0.936	[0.858, 1.022]

"Yes" = 1, "No" = 0. Categorical predictors are dummy-coded against their first group.

Figure 35. The model fit and the coefficients with odds ratios.

What it means for our example

The model uses 366 students and predicts passing better than chance: $\chi^2(4) = 63.56$, $p < .001$, Nagelkerke $R^2 = .225$ (McFadden .141, Tjur .186), AUC = .780, 95% CI [.73, .83] (acceptable). The events-per-variable check (28.0) and the Hosmer-Lemeshow test, $\chi^2(8) = 5.80$, $p = .669$, are fine. Each additional weekly hour of study multiplies the odds of passing by 1.29 (95% CI [1.17, 1.42], $p < .001$) and each additional hour of sleep by 1.62 (95% CI [1.25, 2.10], $p < .001$); the odds of passing for men are 0.40 times those of women (95% CI [0.25, 0.66], $p < .001$), equivalent to saying that women's odds are 2.47 times men's. Anxiety_score has no significant effect, OR = 0.94, 95% CI [0.86, 1.02], $p = .139$.

The classification table shows what the model does with each student: of the 112 who failed, it predicts 44 correctly (39.3%); of the 254 who passed, it predicts 232 correctly (91.3%); overall 75.4% are classified correctly, against 69.4% if the larger group ("Yes") were guessed for everyone. The overall accuracy looks good mostly because most students pass: the model finds failing students poorly. For an outcome like this, read the AUC and the sensitivity for the rarer group, not the overall accuracy alone.

How to report it

"A binary logistic regression predicting passing (Passed = Yes) from Gender, Study_hours_week, Sleep_hours and Anxiety_score was significant, $\chi^2(4, N = 366) = 63.56$, $p < .001$, Nagelkerke $R^2 = .23$, AUC = .78. Men had lower odds of passing than women, OR = 0.40, 95% CI [0.25, 0.66]; each weekly hour of study, OR = 1.29, 95% CI [1.17, 1.42], and each hour of sleep, OR = 1.62, 95% CI [1.25, 2.10], raised the odds."

Exercise: regression on practice_marketing.csv

Open practice_marketing.csv. (a) Run a multiple linear regression of Annual_spend on Income_k, Tenure_months, Loyalty_member, Visits_per_month and Satisfaction_1to10. Answer: $R^2 = .81$, $F(5, 1335) = 1108.12$, $p < .001$ ($n = 1341$); Visits_per_month dominates ($\beta = .86$), followed by Income_k ($\beta = .17$) and Satisfaction_1to10 ($\beta = .08$); Loyalty_member adds a little ($\beta = .04$); Tenure_months is not significant ($p = .747$). (b) Run a binary logistic regression of Churned (event "Yes") on Satisfaction_1to10, Complaints_12m, Tenure_months and Loyalty_member. Answer: AUC = .75 (acceptable); each satisfaction point has OR = 0.61 (95% CI [0.54, 0.68]), each complaint OR = 1.39 (95% CI [1.21, 1.60]) and loyalty membership OR = 0.69 (95% CI [0.49, 0.96]). Which of the checks does the regression flag? (c) Repeat (a) with the logarithms of Annual_spend and Income_k (Section 1.6). Answer: $R^2 = .77$ ($n = 1,268$ complete cases, because the 82 zero-spend customers and the missing incomes drop out); Visits_per_month is again the strongest predictor ($\beta = .84$), followed by Income_k_ln ($\beta = .17$), and the Breusch-Pagan test no longer flags the error variance ($p = .431$).

11. Power analysis and sample size planning

Power analysis answers two questions: how many people do I need? and how likely is my study to find an effect that is there? It needs no data file: click **Power analysis** in the menu.

Concept: errors, power and effect size

A **Type I error** is a false positive (finding an effect that is not there); its probability is the significance level α , usually .05. A **Type II error** is a false negative (missing an effect that is there); its probability is β . **Power** = $1 - \beta$ is the probability of finding an effect that really exists; .80 is the usual target. Power rises with the sample size, the size of the true effect and α . The **effect size** is the smallest effect that would matter to you, or one taken from earlier studies; Cohen's (1988) conventions (small, medium, large) are a fallback, not a substitute for thinking.

11.1 The three kinds of power analysis

1. Which analysis will you run? Choose the test: **Independent-samples t-test (Compare two groups)**, **Paired-samples t-test**, **One-sample t-test**, **One-way ANOVA**, **fixed effects**, **Chi-square test**, **Pearson correlation test**, **Multiple linear regression**, **F-test of R^2** , or the PLS-SEM inverse square root method (not covered here).
2. What do you want to find? **A priori power analysis (The sample size I need)** before data collection, **Post hoc power analysis (The power of a given sample size)** for a given sample, or **Sensitivity power analysis (The smallest effect I can detect)**.
3. Enter the effect size, α (.05), the power (.80) and the details of the design (the number of groups, predictors or degrees of freedom). The result and a ready-made sentence appear immediately (Figure 36).

SaNaML Statistics & Machine Learning · 0.10.2

Home Data Analyses Charts **Power analysis** Clustering Machine learning Word report

▼ What is this screen? What do I do here?

Work out how many respondents you need before collecting data (a priori), how much power a planned or finished study has, or the smallest effect it can detect. No data file is needed. The sentence at the bottom can go straight into your proposal or methods section.

1 · Which analysis will you run?

☒ **Multiple linear regression, F-test of R^2 (also the usual PLS-SEM check)**
For PLS-SEM, use the largest number of arrows pointing at one construct as the number of predictors.

☐ Inverse square root method (PLS-SEM minimum sample size)
Minimum sample size from the smallest path coefficient you expect (Kock & Hadaya, 2018).

☐ Independent-samples t-test (Compare two groups)
Two groups of different people.

☐ Paired-samples t-test (Same people twice)
Before vs after, or two ratings per person.

☐ One-sample t-test (One score against a fixed value)
Compares one mean with a fixed value.

☐ One-way ANOVA, fixed effects (Compare three or more groups)
One-way ANOVA, effect size f.

☐ Chi-square test (Goodness of fit or independence)
Goodness of fit or independence; effect size w.

☐ Pearson correlation test (Is a correlation different from zero?)
Test whether a correlation differs from zero.

2 · What do you want to find?

☒ **A priori power analysis (The sample size I need)**
Most common in proposals.

Figure 36. The power analysis screen as it opens: a regression with four predictors needs $N = 85$.

The effect-size measure depends on the test: Cohen's d (t-tests; d_z for paired data), f (ANOVA), f^2 (regression), w (chi-square) and r (correlation). Table 5 shows how to convert the effect sizes found in the earlier chapters.

Table 5

Effect sizes used for planning, and how they were obtained from the results of this manual

Planned analysis	Effect size input	From
Independent-samples t-test	Cohen's $d = 0.44$	Chapter 5, Exam_score by Gender
One-way ANOVA, 3 groups	Cohen's $f = 0.24$	Chapter 6: $f = \sqrt{(\eta^2 / (1 - \eta^2))} = \sqrt{(.054 / .946)} = 0.24$
Chi-square, 2 df	Cohen's $w = 0.17$	Chapter 9: $w = \text{Cramér's } V \times \sqrt{(\text{smaller table dimension} - 1)}$, here $V = .17$ for the 3×2 table, because the smaller dimension is 2
Pearson correlation	$r = .27$	Chapter 10, Study_hours_week and Exam_score
Multiple regression, 4 predictors	$f^2 = 0.21$	Chapter 10: $f^2 = R^2 / (1 - R^2) = .172 / .828 = 0.21$
Paired-samples t-test	$d_z = 0.50$	A medium effect, as a convention

11.2 A priori: how many students do I need?

Suppose you want to repeat the comparison of women and men ($d = 0.44$) in a new study, with a two-sided test, $\alpha = .05$ and power .80. Choose **Independent-samples t-test, A priori** and type 0.44 for the effect size. SaNaML reports $N = 166$, 83 per group (achieved power .805).

The other analyses in Table Table 5 give: one-way ANOVA with 3 groups and $f = 0.24$, $N = 171$ (57 per group, achieved power .802); chi-square with 2 degrees of freedom and $w = 0.17$, $N = 334$ (.801); a correlation of $r = .27$, $N = 105$ (.802); a regression with 4 predictors and $f^2 = 0.21$, $N = 62$ (.802); and a paired-samples t-test with $d_z = 0.50$, $N = 34$ (.808). A medium effect ($f^2 = 0.15$) with four predictors needs $N = 85$ (Figure 36). A one-sided test of $d = 0.50$ needs $N = 102$ (51 per group, achieved power .806) because the test is concentrated in one direction; use it only with a directional hypothesis made in advance.

How to report it

"An a priori power analysis for an independent-samples t-test indicated that 166 participants (83 per group) are needed to detect Cohen's $d = 0.44$ with $\alpha = .05$ (two-sided) and power = .80 (achieved power .805)."

Tip: plan for drop-out and missing data

The numbers SaNaML gives are the sample you need *after* exclusions. If you expect 5% of your respondents to drop out, give inaccurate answers or leave variables blank (as in this practice file), recruit the target divided by .95. Also remember that SaNaML's power analyses assume the standard conditions of each test; the sample size for a multilevel or non-random sample may need more.

11.3 Sensitivity: what can I detect with the students I have?

With $N = 400$ and power .80, which effects can a study detect? Choose **Sensitivity power analysis** and type the sample size. With four predictors, a regression can detect $f^2 = 0.030$ or larger; an independent-samples t-test can detect Cohen's $d = 0.281$ or larger. These are small effects: with 400 students almost anything of a practical size will be found, and a non-significant result is informative.

11.4 Post hoc power: use with care

Post hoc power analysis computes the power of a given sample size for an assumed effect: with $N = 387$ and $d = 0.44$, an independent-samples t-test has a power of .991. Use it to *plan* (what could a study of this size detect?), not to explain a result after the fact. Power computed from the effect size observed in the same data is a function of the p-value and adds no information; a non-significant result is better described by its confidence interval (Hoenig & Heisey, 2001).

Exercise: plan a study from the marketing data

In `practice_marketing.csv` the new offer was answered by 44% of customers and the standard offer by 26%, Cramér's $V = .19$ (chi-square test, $df = 1$, $N = 1500$). Use $w = 0.19$ and 1 degree of freedom to find the sample size a new test of this effect needs, then the sensitivity of a study with 200 customers. Use the paired result of chapter 5 ($d_z = 0.74$) to find how many customers a before-and-after comparison needs.

12. Machine learning as a check on regression

Linear regression assumes that each predictor acts in a straight line and that predictors do not interact. The Machine learning screen of SaNaML tests that assumption from the outside: it fits flexible models that need no such assumption and asks whether any of them predicts *new* students better than the linear model does. This chapter is a short introduction for that single purpose. It is not a course in machine learning, and a flexible model that predicts well does not tell you why.

Concept: out-of-sample prediction and cross-validation

A model always fits the data it was built on better than it will fit new data, because it partly learns the noise (overfitting). The honest measure is *out-of-sample* performance: fit the model on most of the students, predict the rest, and repeat. In 10-fold cross-validation the students are divided into 10 parts; each part is predicted in turn by a model fitted to the other nine. Repeating this 5 times with different divisions makes the result stable. **RMSE** is the typical size of the prediction error in the units of the outcome (lower is better); **R²** here is the share of variance predicted in the unseen students. SaNaML repeats every data-dependent step inside each training set, so nothing leaks from the test rows (nested cross-validation).

12.1 Set up the comparison

1. Click **Machine learning** in the menu.
2. In **Outcome to predict**, choose **Exam_score** (a score, so the task is regression).
3. Under **Predictors**, click **Select none** and tick **Gender**, **Study_hours_week**, **Sleep_hours** and **Anxiety_score**, the same four predictors as the regression of Section 10.5.

SaNaML reports what it did with missing data: the 13 students without an Exam_score are left out, leaving 387, and the other missing values are filled in within each training fold. It also protects you from a respondent identifier: StudentID is offered in the list of predictors but is unticked by default; if you tick it, SaNaML shows an *attention* message, "StudentID looks like a respondent identifier (every value different). A model could learn the order in which data were collected, or memorise individual respondents, rather than anything general about them. It is always left out of the model, whether or not its box is ticked." Card 2 sets the validation plan: the recommended plan for 387 respondents is 10-fold cross-validation repeated 5 times (the quick 5-fold plan is for trying things out, the thorough plan for a result you will report), and the **Random seed** makes the splits reproducible. Card 3 lists the models: a null model that always predicts the average, linear regression, elastic net, a decision tree (CART), a random forest and two boosting methods, XGBoost and LightGBM; a stacked ensemble can be added.

Click **Run the comparison**. The run took under two minutes on the practice file.

12.2 Reading the comparison

Model	RMSE M (SD)	95% CI	MAE	R ²	Q ² predict	Δ vs best (p)
LightGBM	8.279 (0.886)	[7.633, 8.926]	6.702	.247	.267	best
XGBoost	8.294 (0.917)	[7.625, 8.964]	6.721	.245	.265	0.015 (.821)
Random forest	8.449 (0.919)	[7.778, 9.120]	6.908	.219	.240	0.169 (.165)
Decision tree (CART)	8.684 (0.960)	[7.983, 9.385]	7.041	.172	.195	0.404 (.055)
Elastic net	9.012 (1.120)	[8.194, 9.831]	7.202	.116	.141	0.733 (.013)
Linear regression	9.021 (1.186)	[8.155, 9.887]	7.182	.112	.136	0.741 (.019)
Null model	9.766 (1.077)	[8.980, 10.552]	7.887	-.031	.000	1.487 (< .001)

RMSE M (SD) over 50 test folds; 95% CI with the Nadeau-Bengio corrected variance. Δ vs best: mean per-fold difference from the best model, corrected resampled t-test p.

Figure 37. Out-of-sample performance of each model.

Table Table 6 gives the main numbers. The null model, which ignores the predictors, has an R² of $-.031$: any useful model must beat it. Linear regression explains 11.2% of the variance in students it has not seen (RMSE = 9.021). LightGBM does best, RMSE = 8.279, 95% CI [7.633, 8.926], R² = .247, and its advantage over linear regression is statistically significant, Δ RMSE = -0.741 , $p = .019$ (the corrected resampled t-test, which allows for the overlap between training sets; Nadeau & Bengio, 2003).

Table 6

Out-of-sample prediction of Exam_score (50 test folds; lower RMSE is better)

Model	RMSE (SD)	R ²	Difference from best	p
LightGBM	8.279 (0.886)	.247	best	—
XGBoost	8.294 (0.917)	.245	0.015	.821
Random forest	8.449 (0.919)	.219	0.169	.165
Decision tree (CART)	8.684 (0.960)	.172	0.404	.055
Elastic net	9.012 (1.120)	.116	0.733	.013
Linear regression	9.021 (1.186)	.112	0.741	.019
Null model	9.766 (1.077)	$-.031$	1.487	< .001

What it means for our example

Two things follow. First, the regression of Section 10.5 explained 17.2% of the variance of Exam_score in the sample it was fitted to, but only 11.2% in students it had not seen: the in-sample figure is optimistic, and the out-of-sample one is the honest estimate. Second, the boosting methods and the random forest predict better than the straight-line model, which suggests that part of the relationship may not be a straight line. That is consistent with chapter 10, where the rank correlation of Study_hours_week with Exam_score was larger than the Pearson correlation. The gain is modest in absolute terms: even the best model leaves about three-quarters of the variance unexplained.

12.3 Which predictors matter, and how?

Flexible models are opaque, so SaNaML explains the best model with three tools. **SHAP values** give each predictor's average contribution to the predictions, in points of the outcome (Figure 38).

Permutation importance shuffles one predictor in the test rows and measures how much the error rises. **Accumulated local effects (ALE)** show how the prediction changes as one predictor changes, allowing for correlated predictors (Apley & Zhu, 2020).

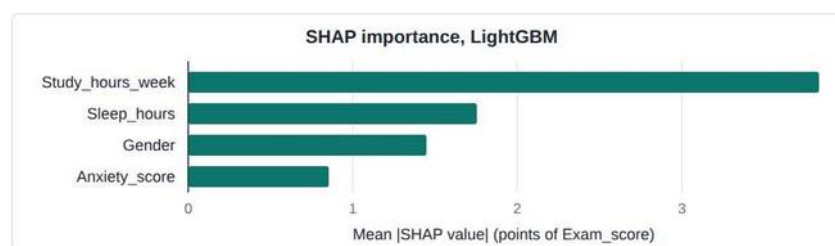


Figure 38. The SHAP importance of each predictor in the LightGBM model.

Study_hours_week is the strongest driver (mean $|\text{SHAP}| = 3.834$ points, permutation importance 2.402), followed by Sleep_hours (1.754; 0.513), Gender (1.448; 0.298) and Anxiety_score (0.855; 0.007). Anxiety_score adds almost nothing to prediction on unseen students, which matches its non-significant coefficient in the regression ($\beta = -.08$, $p = .121$). The two rankings agree, and the other models rank the predictors similarly; the linear model ranks Study_hours_week third, (Spearman's ρ between the LightGBM and linear-regression rankings = .40), possibly because of the extreme study times noted in chapter 10. Importance describes what the model uses; it is not evidence of a causal effect (Molnar, 2022).

The ALE plots show how the prediction moves. For Study_hours_week the effect is about -8 points at the lowest values, close to zero between 6 and 8 hours, and about $+10$ points at the longest times; for Gender the average prediction is 56.10 for women and 53.03 for men.

Note: read the plot as well as the sentence

SaNaML writes a one-sentence summary of each ALE curve. For Study_hours_week it says that the relationship "is not strictly monotonic: the prediction rises overall (from -8.443 to 10.358) with a small reversal along the way", and for Sleep_hours that "the prediction rises to a peak near 9.70 and then falls". The sentences are still brief: the last interval of the Study_hours_week curve spans 10.7 to 60 hours, so it summarises the few extreme study times in one step. Use the table and the plot to describe the shape.

12.4 Calibration

For a score outcome, SaNaML also checks calibration: do the predictions have the right scale? Regressing the actual Exam_score on the cross-validated predictions of LightGBM gives a slope of 0.93, 95% CI [0.78, 1.08], and the mean prediction error is -0.011 points. A slope near 1 means the predictions are about as spread out as they should be; a slope well below 1 would be a sign of overfitting.

How to report it

"Seven prediction models were compared with 10-fold cross-validation repeated 5 times (50 train/test splits), with every model tuned by an inner 5-fold cross-validation inside each training set (nested cross-validation; seed 2026); missing values were imputed and predictors standardised within each training fold. LightGBM performed best (RMSE = 8.279, 95% CI [7.633, 8.926]; $R^2 = .247$), significantly lower than linear regression, $\Delta\text{RMSE} = -0.741$, corrected resampled $t(49) = -2.42$, $p = .019$."

12.5 A yes/no outcome: cut-offs, precision and recall

The same screen predicts a yes/no outcome. Choose **Passed** as **Outcome to predict**; SaNaML asks **Which answer is the event (coded 1)?**, so choose **Yes**. Tick the same four predictors (Gender, Study_hours_week, Sleep_hours and Anxiety_score; never Exam_score, from which Passed was made) and choose the **Quick look** plan, 5-fold cross-validation without repeats, with only the null model, logistic regression and the random forest (Figure 39). The quick plan is fast but noisy; use it to try things out, not to report.

Out-of-sample performance (nested cross-validation)								
Model	AUC M (SD)	95% CI	Accuracy	Balanced accuracy	F1	MCC	Brier score	Δ vs best (p)
Logistic regression	.782 (.060)	[.671, .892]	.750	.647	.835	.357	0.170	best
Random forest	.748 (.031)	[.690, .806]	.737	.661	.819	.350	0.183	-.034 (.500)
Null model	.500 (.000)	[.500, .500]	.695	.500	.820	.000	0.212	-.282 (.002)

AUC M (SD) over 5 test folds; 95% CI with the Nadeau-Bengio corrected variance. Δ vs best: mean per-fold difference from the best model, corrected resampled t-test p.

Figure 39. Out-of-sample performance of the three models predicting Passed = Yes (quick look).

Logistic regression is the best model: AUC = .782 (SD = .060), 95% CI [.671, .892], accuracy = .750 and balanced accuracy = .647. The random forest is not significantly different (AUC = .748, difference -.034, $p = .500$) and the null model, with an AUC of .500, is clearly worse ($p = .002$). The interval is wide because there are only 5 test folds. SaNaML adds that flexible models often need many more respondents than linear models before they pay off, and with 400 students a flexible model does not beat the logistic regression here.

A model gives each student a *probability* of passing. To call a student a case you need a **cut-off**. The section *Results · Cut-offs, precision and recall* (Figure 40) has a number box, **Cut-off (probability, between 0 and 1)**; type a value and everything updates. At 0.50, 82% of the students are flagged; of the students who really passed, 91% are found (**sensitivity**, also called **recall**) and of those flagged, 77% really passed (**precision**). The count table gives the same story in numbers: 253 passers correctly called, 25 missed, 75 failers wrongly flagged and 47 correctly left alone. **Specificity** (failers correctly left alone, 47 of 122) is only 0.39. The table *Measures at every cut-off* and the chart (Figure 41) show how the measures trade off. Raising the cut-off to 0.70 flags 56% of the students, finds 70% of the passers and raises the precision to 87%, with a specificity of 0.75.

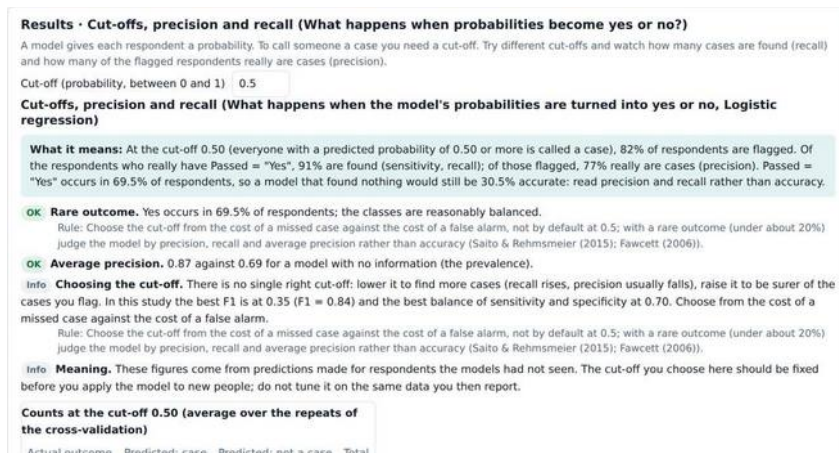


Figure 40. The cut-off box and the summary of the results at a cut-off of 0.50.

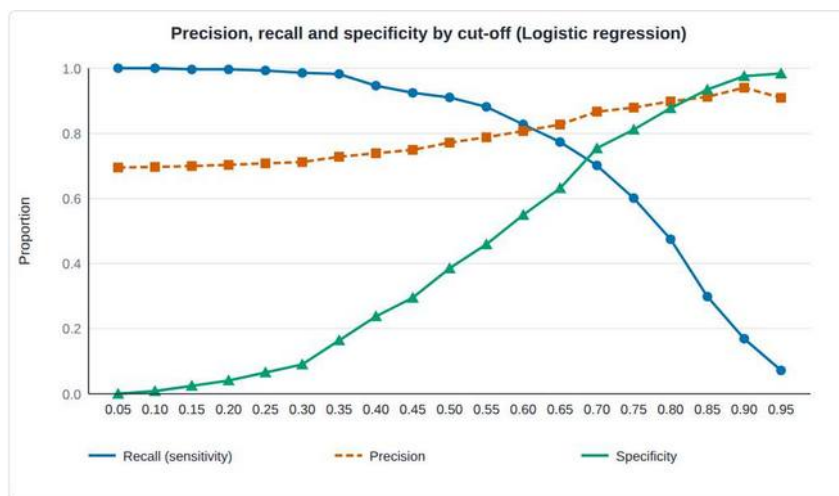


Figure 41. Precision, recall and specificity at every cut-off.

SaNaML also reports the average precision, the area under the precision–recall curve: 0.87, against 0.69 for a model with no information (the prevalence). It checks whether the outcome is rare: passing occurs in 69.5% of the students, so the classes are reasonably balanced; with a rare outcome (under about 20%) accuracy misleads and you should judge a model by precision, recall and average precision (Saito & Rehmsmeier, 2015). In this study the best F1 is at 0.35 (F1 = 0.84) and the best balance of sensitivity and specificity at 0.70. There is no single right cut-off: choose it from the cost of a missed case against the cost of a false alarm (Fawcett, 2006), fix it before you apply the model to new students, and do not tune it on the data you then report.

How to report it

"On cross-validated predictions of Passed = "Yes", logistic regression at a cut-off of 0.70 had sensitivity = 0.70, specificity = 0.75, precision = 0.87 and F1 = 0.78; the average precision (area under the precision–recall curve) was 0.87, against 0.69 expected by chance (the prevalence; Saito & Rehmsmeier, 2015)."

Exercise: machine learning on another file

In practice_marketing.csv, predict Annual_spend from Income_k, Tenure_months, Loyalty_member, Visits_per_month and Satisfaction_1to10. In the regression of the exercise in chapter 10 these predictors explain 81% of the variance in the sample ($R^2 = .81$). Compare that figure with the out-of-sample R^2 of linear regression, then with that of the best model, and look at the ALE plot of Visits_per_month, the strongest predictor in the regression. Do the flexible models gain over the linear model?

13. Reporting your results

Statistics are only useful when other people can read them. This chapter shows how SaNaML collects results into a Word report and gives a short checklist for reporting tests in APA style.

13.1 Collecting results in the Word report

Every result in SaNaML (an analysis, a chart or a power analysis) ends with the sentence that you would put in your text, with two buttons: **Copy this sentence** and **Add to Word report** (Figure 42). Adding a result marks it with the label *In the Word report* and a link to the report. Add the results as you go; the report keeps them in the order added.

How to report it (APA style): An independent-samples Welch's t-test showed that Exam_score differed between female (M = 56.57, SD = 9.64) and male (M = 52.38, SD = 9.57), $t(313.32) = 4.17$, $p < .001$, $d = 0.44$, 95% CI [0.23, 0.64].

Copy this sentence

Add to Word report

Figure 42. The APA sentence with Copy this sentence and Add to Word report.

Click **Word report** in the menu to open the collected results (Figure 43). Each entry shows its title and the interpretation sentence, with buttons to move it up or down or remove it, and **Report title** sets the heading of the document. **Download Word report (.docx)** creates the file, named after the title (here "Common statistical tests practice students (SaNaML).docx").

Word report (5 results)

Independent-samples t-test (Comparing Exam_score between female and male)
female scored higher than male on Exam_score (difference 4.19 points). The difference is statistically significant and its size is small ($|d| = 0.44$).

↑ ↓ Remove

One-way ANOVA (Comparing Post_score across Teaching_method)
Post_score differs across the 3 Teaching_method groups (small effect, $\omega^2 = .049$). Pairs that differ: Blended vs Lecture; Flipped vs Lecture.

↑ ↓ Remove

Paired-samples t-test (Comparing Pre_score and Post_score, same respondents)
Post_score is higher than Pre_score by 6.72 points on average. The change is statistically significant and large ($d_z = 1.06$).

↑ ↓ Remove

Chi-square test of independence (Teaching_method by Passed)
Teaching_method and Passed are related (Cramér's $V = .17$, small). The cells with large adjusted residuals (beyond ± 1.96) show where the pattern comes from.

↑ ↓ Remove

Means with confidence intervals of Exam_score by Teaching_method (Do the group averages differ?)
Mean Exam_score: Blended 56.72 [54.97, 58.48]; Flipped 55.32 [53.70, 56.94]; Lecture 53.12 [51.43, 54.81]. Where two intervals do not overlap, the groups very likely differ; overlapping intervals do not by themselves show that they are the same. Run the comparison in the Analyses to test it.

↑ ↓ Remove

Report title: statistical tests: practice students

Download Word report (.docx)

Figure 43. The Word report screen with five results, the report title and the download button.

The document for the five results of this manual's practice session (an independent-samples t-test, a one-way ANOVA, a paired-samples t-test, a chi-square test and a chart of means) contains, in this order:

- the report title, and a line stating that the analyses were run in SaNaML 0.10.2 on the date of the download and that all computations were done on the analyst's own computer;
- for each result, a heading, the APA sentence, an *Interpretation*, the *Checks* that were run, the numbered tables (descriptives, the tests, the effect size with its confidence interval, the rank-based alternative and the notes under each table) and, for a chart, the figure with its caption;

- a table of *Criteria used*, listing each rule of thumb that was applied in the report and its source;
- the *References* in APA style, in alphabetical order.

Tip: check before you submit

The sentences and tables are a starting point. Read each against your design: the sentence names the groups in the file's own wording, a one-tailed test is marked and worded as a direction ("was higher for A than for B"), and the sentence cannot know which hypotheses you set in advance. Edit the document in your word processor, keep the numbers as SaNaML gives them and add the context that only you have: the sample, the design and what the result means for your question.

13.2 A reporting checklist

1. **Describe the data.** State the sample size, the population, how the sample was obtained and whether the data are real or simulated. Give the descriptive statistics (M and SD, or the median and range for skewed variables) for each variable analysed.
2. **Report how you handled problems.** Say how many respondents were excluded and why (Chapter 2), how missing values were treated (each analysis states its n) and whether scales were reversed or combined and which variables were transformed or recoded (Sections 1.5 and 1.6).
3. **Name the test and the reason.** For example, "Welch's t-test, because the group sizes were unequal".
4. **Report the statistic, the degrees of freedom, the exact p-value and an effect size with its confidence interval.** For example, $t(313.32) = 4.17$, $p < .001$, $d = 0.44$, 95% CI [0.23, 0.64].
5. **Report adjustments.** Say which correction was used for the follow-up tests (Tukey-Kramer, Games-Howell, Dunn or Holm) and for the family of tests, and report how many tests you ran.
6. **Interpret the size and check the assumptions.** Use the benchmarks (Appendix B) as a guide, judge importance in the context of your field, and say in one sentence which checks were made and which alternative was used when one failed.

Cut-off values: formatting numbers in APA style

Report p to three decimals and write " $p < .001$ " for smaller values. Leave out the leading zero for values that cannot exceed 1 (p , r , V , η^2 , ω^2 , R^2 , α) and keep it for d , g , f and B . Italicise the statistical symbols in the final document. SaNaML's sentences and tables already follow this style. Use the same number of decimals across a table.

Note: simulated data

The results in this manual come from a simulated file made for practice. The p-values, effect sizes and conclusions describe the simulated students only and must not be quoted as findings about real students. When you report your own data, replace every example with your own output.

Exercise: build a report

Open practice_marketing.csv. Run (a) the independent-samples t-test of Annual_spend between the two Loyalty_member groups, (b) the chi-square test of Campaign_group by Responded and (c) the multiple linear regression of Annual_spend of chapter 10. Add the three to the Word report, give the report a title, move the chi-square test to the first place and download the document. Check that the chi-square result in the file reads $\chi^2(1, N = 1500) = 54.50, p < .001$, Cramér's V = .19.

Appendix A. Codebook of `practice_students.csv`

The file holds 400 simulated students. The values were generated by a computer program; there are no real persons in it. Table 7 lists every variable with its type in SaNaML, its values or range and the number of missing values.

Table 7

The variables of `practice_students.csv` (N = 400)

Variable	Type	Values or range	Missing
StudentID	label	ST001 to ST400; a label only	0
Gender	categorical	female (246), male (154)	0
Age	scale	17 to 24 years; M = 20.23, SD = 1.40	0
Year	scale	1 to 4 (treated as categorical in Chapter 1 where needed); M = 2.38	0
Programme	categorical	Arts (92), Business (163), Science (145)	0
Residence	categorical	Home (195), Hostel (119), Rented (74)	12
Teaching_method	categorical	Blended (129), Flipped (129), Lecture (142)	0
Study_hours_week	scale	0.7 to 60; M = 7.28, SD = 5.80	0
Sleep_hours	scale	3.5 to 9.9; M = 6.68, SD = 1.07	12
Anxiety_score	scale	7.1 to 26.2; M = 13.50, SD = 2.90	22
Pre_score	scale	25.3 to 95.0; M = 59.37, SD = 11.65	0
Post_score	scale	28.3 to 100.0; M = 66.09, SD = 13.47	0
Score_T1, Score_T2, Score_T3	scale	Three measurements of the same test; M = 59.37, 62.55 and 66.09	0
Exam_score	scale	29.0 to 80.1; M = 54.97, SD = 9.81	13
Project_score	scale	32.2 to 99.2; M = 59.85, SD = 12.02	0
Passed	categorical	Yes (278), No (122); the pass/fail version of Exam_score, pass mark 50	0
Pass_before	categorical	Yes (319), No (81); the pass/fail version of Pre_score	0
Pass_after	categorical	Yes (358), No (42); the pass/fail version of Post_score	0
Satisfaction_1to5	scale	One rating, 1 to 5; M = 3.05, SD = 1.00	16
Mot1 to Mot6	scale	Motivation items, 1 to 5	0
Anx1 to Anx6	scale	Anxiety items, 1 to 5; Anx3 and Anx5 are worded in the opposite direction	0

Exam_score is missing for 13 students, but Passed is filled in for all of them (11 Yes and 2 No), so Passed has no missing values although Exam_score has. For analyses that involve Passed, N is 400; for analyses that involve Exam_score, at most 387.

Appendix B. Assumptions and benchmarks of the tests

Table 8

The tests of this manual: assumptions, effect sizes and benchmarks

Test	Main assumptions	Effect size and benchmarks
Independent-samples t (Welch)	Independent observations; roughly normal sample means (about 30 or more per group is enough); Welch needs no equal variances	Cohen's d: .2 small, .5 medium, .8 large
Paired-samples t	Independent pairs; roughly normal differences	d _z , same benchmarks
One-sample t	Independent observations; roughly normal sample mean	d, same benchmarks
Mann–Whitney U, Wilcoxon	Independent cases or pairs; similar shapes if medians are compared	Rank-biserial r: .1, .3, .5
One-way ANOVA	Independent observations; roughly normal means; equal variances (otherwise Welch and Games–Howell)	η^2 , ω^2 : .01, .06, .14
Kruskal–Wallis	Independent observations; similar shapes	ϵ^2
Repeated-measures ANOVA	Sphericity (Mauchly); if violated, Greenhouse–Geisser or Huynh–Feldt	Partial η^2
Factorial ANOVA, ANCOVA	As ANOVA; for ANCOVA, parallel slopes of the covariate in all groups	Partial η^2
MANOVA	Independent observations; multivariate normality; similar covariance matrices	Pillai's V; partial η^2 for each follow-up ANOVA
Chi-square	Independent observations; expected counts not below 1 and at most 20% below 5	Cramér's V (ϕ for 2×2): .1, .3, .5; w: .1, .3, .5
McNemar	Paired yes/no answers	Percentage change; exact test if few change
Pearson r	Linear relationship; no extreme values	r: .1, .3, .5
Spearman ρ , Kendall τ -b	Monotonic relationship	ρ and τ read as r
Linear regression	Linear effects; independent, normal errors with constant spread; no strong collinearity (VIF below 5); no influential cases	R^2 ; f^2 : .02, .15, .35
Logistic regression	Independent observations; at least 10 cases in the smaller outcome group per coefficient	Odds ratios; Nagelkerke R^2 ; AUC: .7 acceptable, .8 excellent, .9 outstanding

The rules behind the checks are: Welch's t-test by default (Delacre et al., 2017); about 30 or more cases per group before non-normality stops mattering (Lumley et al., 2002); Levene's test for equal variances (Levene, 1960), with Games–Howell comparisons when it fails (Games & Howell, 1976); Greenhouse–Geisser when $\epsilon < .75$ and Huynh–Feldt otherwise (Girden, 1992); expected counts of at least 1, and at most 20% below 5, in a chi-square table (Cochran, 1954); at least 50 + 8k respondents

for k regression predictors (Green, 1991); VIF below 5 (Hair et al., 2019; Kutner et al., 2005); Shapiro–Wilk for the residuals (Shapiro & Wilk, 1965); Breusch–Pagan (Koenker) for constant error variance (Breusch & Pagan, 1979; Koenker, 1981); Cook's distance below 1 (Cook & Weisberg, 1982); at least 10 cases in the smaller outcome group per coefficient (Peduzzi et al., 1996); and AUC, Hosmer–Lemeshow (Hosmer et al., 2013). The benchmarks are Cohen's (1988) conventions as SaNaML applies them; they are rules of thumb, not laws, and the importance of an effect depends on the field. For Kruskal–Wallis, partial η^2 and Pillai's V , SaNaML reports the value and the other statistics, and its screens do not print a benchmark for all of them, so use the field's own conventions.

Appendix C. Glossary

Table 9

Terms used in this manual

Term	Meaning
Adjusted p-value	A p-value corrected for the number of tests in the family (Holm, Tukey-Kramer, Games-Howell and others).
Alpha (α)	The significance level, the false-positive rate you accept; also Cronbach's α , a reliability coefficient (the context shows which).
ANCOVA	ANOVA with a covariate, which adjusts the group means for a related scale variable.
Calibration	Whether predictions have the right scale; a calibration slope of 1 is ideal.
Confidence interval	A range of values compatible with the data at the stated confidence (here 95%).
Covariate	A scale variable that is related to the outcome and is held constant in an ANCOVA or regression.
Cramér's V	Effect size for a table of counts, from 0 to 1.
Cut-off (probability)	The predicted probability from which a respondent is called a case; it decides sensitivity, specificity and precision.
Effect size	The size of a difference or relationship, independent of the sample size.
Estimated marginal mean	A group mean adjusted for the other terms of the model.
Holm adjustment	A step-down correction for multiple tests that is never less powerful than Bonferroni's.
Indicator variable	A variable that is 1 for a chosen category and 0 for all others.
Interaction	An effect of one factor that depends on the level of another.
Levene's test	A test of equal variances in groups.
Median split	Two groups made at the median of a scale variable; a value equal to the median goes to the lower group.
Missing completely at random	Missingness that is unrelated to the data (Little's test).
Natural logarithm	$\ln(x)$; a transformation that pulls in a long right tail and needs positive values.
Odds ratio	The multiplicative change in the odds of an event for one unit of a predictor.
Power	The probability of finding an effect that really exists.
Precision	The share of the respondents flagged as cases who really are cases.
Sensitivity (recall)	The share of the real cases that the model finds.
Specificity	The share of the non-cases that the model correctly leaves alone.
Sphericity	The assumption that the differences between all pairs of repeated measures have equal variance.
Straight-lining	Giving the same answer to every item of a questionnaire.
Type III sums of squares	Each effect is tested after all the others; the usual choice for unbalanced factorial designs.
VIF	Variance inflation factor, a measure of collinearity among predictors.

References

- Apley, D. W., & Zhu, J. (2020). Visualizing the effects of predictor variables in black box supervised learning models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 82(4), 1059–1086. <https://doi.org/10.1111/rssb.12377>
- Bakeman, R. (2005). Recommended effect size statistics for repeated measures designs. *Behavior Research Methods*, 37(3), 379–384. <https://doi.org/10.3758/BF03192707>
- Bouckaert, R. R., & Frank, E. (2004). Evaluating the replicability of significance tests for comparing learning algorithms. In H. Dai, R. Srikant, & C. Zhang (Eds.), *Advances in knowledge discovery and data mining* (pp. 3–12). Springer. https://doi.org/10.1007/978-3-540-24775-3_3
- Breusch, T. S., & Pagan, A. R. (1979). A simple test for heteroscedasticity and random coefficient variation. *Econometrica*, 47(5), 1287–1294. <https://doi.org/10.2307/1911963>
- Brown, M. B., & Forsythe, A. B. (1974). Robust tests for the equality of variances. *Journal of the American Statistical Association*, 69(346), 364–367. <https://doi.org/10.1080/01621459.1974.10482955>
- Cochran, W. G. (1954). Some methods for strengthening the common χ^2 tests. *Biometrics*, 10(4), 417–451. <https://doi.org/10.2307/3001616>
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum.
- Cook, R. D., & Weisberg, S. (1982). *Residuals and influence in regression*. Chapman & Hall.
- Cronbach, L. J. (1951). Coefficient alpha and the internal structure of tests. *Psychometrika*, 16(3), 297–334. <https://doi.org/10.1007/BF02310555>
- Curran, P. G. (2016). Methods for the detection of carelessly invalid responses in survey data. *Journal of Experimental Social Psychology*, 66, 4–19. <https://doi.org/10.1016/j.jesp.2015.07.006>
- Delacre, M., Lakens, D., & Leys, C. (2017). Why psychologists should by default use Welch's t-test instead of Student's t-test. *International Review of Social Psychology*, 30(1), 92–101. <https://doi.org/10.5334/irsp.82>
- Dunn, O. J. (1964). Multiple comparisons using rank sums. *Technometrics*, 6(3), 241–252. <https://doi.org/10.1080/00401706.1964.10490181>
- Durbin, J., & Watson, G. S. (1950). Testing for serial correlation in least squares regression. I. *Biometrika*, 37(3/4), 409–428. <https://doi.org/10.2307/2332391>
- Fawcett, T. (2006). An introduction to ROC analysis. *Pattern Recognition Letters*, 27(8), 861–874. <https://doi.org/10.1016/j.patrec.2005.10.010>
- Games, P. A., & Howell, J. F. (1976). Pairwise multiple comparison procedures with unequal n's and/or variances: A Monte Carlo study. *Journal of Educational Statistics*, 1(2), 113–125. <https://doi.org/10.3102/10769986001002113>
- Girden, E. R. (1992). *ANOVA: Repeated measures*. Sage.
- Green, S. B. (1991). How many subjects does it take to do a regression analysis? *Multivariate Behavioral Research*, 26(3), 499–510. https://doi.org/10.1207/s15327906mbr2603_7
- Greenhouse, S. W., & Geisser, S. (1959). On methods in the analysis of profile data. *Psychometrika*, 24(2), 95–112. <https://doi.org/10.1007/BF02289823>
- Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2019). *Multivariate data analysis* (8th ed.). Cengage Learning.
- Hoenig, J. M., & Heisey, D. M. (2001). The abuse of power: The pervasive fallacy of power calculations for data analysis. *The American Statistician*, 55(1), 19–24. <https://doi.org/10.1198/000313001300339897>

- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.
- Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied logistic regression* (3rd ed.). Wiley.
- Huynh, H., & Feldt, L. S. (1976). Estimation of the Box correction for degrees of freedom from sample data in randomized block and split-plot designs. *Journal of Educational Statistics*, 1(1), 69–82.
<https://doi.org/10.3102/10769986001001069>
- Kerby, D. S. (2014). The simple difference formula: An approach to teaching nonparametric correlation. *Comprehensive Psychology*, 3, 11.IT.3.1. <https://doi.org/10.2466/11.IT.3.1>
- Koenker, R. (1981). A note on studentizing a test for heteroscedasticity. *Journal of Econometrics*, 17(1), 107–112. [https://doi.org/10.1016/0304-4076\(81\)90062-2](https://doi.org/10.1016/0304-4076(81)90062-2)
- Kramer, C. Y. (1956). Extension of multiple range tests to group means with unequal numbers of replications. *Biometrics*, 12(3), 307–310. <https://doi.org/10.2307/3001469>
- Kruskal, W. H., & Wallis, W. A. (1952). Use of ranks in one-criterion variance analysis. *Journal of the American Statistical Association*, 47(260), 583–621. <https://doi.org/10.1080/01621459.1952.10483441>
- Kutner, M. H., Nachtsheim, C. J., Neter, J., & Li, W. (2005). *Applied linear statistical models* (5th ed.). McGraw-Hill/Irwin.
- Lakens, D. (2013). Calculating and reporting effect sizes to facilitate cumulative science: A practical primer for t-tests and ANOVAs. *Frontiers in Psychology*, 4, 863. <https://doi.org/10.3389/fpsyg.2013.00863>
- Levene, H. (1960). Robust tests for equality of variances. In I. Olkin (Ed.), *Contributions to probability and statistics* (pp. 278–292). Stanford University Press.
- Little, R. J. A. (1988). A test of missing completely at random for multivariate data with missing values. *Journal of the American Statistical Association*, 83(404), 1198–1202.
<https://doi.org/10.1080/01621459.1988.10478722>
- Lumley, T., Diehr, P., Emerson, S., & Chen, L. (2002). The importance of the normality assumption in large public health data sets. *Annual Review of Public Health*, 23, 151–169.
<https://doi.org/10.1146/annurev.publhealth.23.100901.140546>
- Mann, H. B., & Whitney, D. R. (1947). On a test of whether one of two random variables is stochastically larger than the other. *The Annals of Mathematical Statistics*, 18(1), 50–60.
<https://doi.org/10.1214/aoms/1177730491>
- Mauchly, J. W. (1940). Significance test for sphericity of a normal n-variate distribution. *The Annals of Mathematical Statistics*, 11(2), 204–209. <https://doi.org/10.1214/aoms/1177731915>
- Maxwell, S. E., Delaney, H. D., & Kelley, K. (2018). *Designing experiments and analyzing data: A model comparison perspective* (3rd ed.). Routledge.
- Meade, A. W., & Craig, S. B. (2012). Identifying careless responses in survey data. *Psychological Methods*, 17(3), 437–455. <https://doi.org/10.1037/a0028085>
- Molnar, C. (2022). *Interpretable machine learning: A guide for making black box models explainable* (2nd ed.). Independently published.
- Nadeau, C., & Bengio, Y. (2003). Inference for the generalization error. *Machine Learning*, 52(3), 239–281.
<https://doi.org/10.1023/A:1024068626366>
- Nunnally, J. C. (1978). *Psychometric theory* (2nd ed.). McGraw-Hill.
- Olson, C. L. (1976). On choosing a test statistic in multivariate analysis of variance. *Psychological Bulletin*, 83(4), 579–586. <https://doi.org/10.1037/0033-2909.83.4.579>

- Peduzzi, P., Concato, J., Kemper, E., Holford, T. R., & Feinstein, A. R. (1996). A simulation study of the number of events per variable in logistic regression analysis. *Journal of Clinical Epidemiology*, 49(12), 1373–1379. [https://doi.org/10.1016/S0895-4356\(96\)00236-3](https://doi.org/10.1016/S0895-4356(96)00236-3)
- Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3), e0118432. <https://doi.org/10.1371/journal.pone.0118432>
- Shapiro, S. S., & Wilk, M. B. (1965). An analysis of variance test for normality (complete samples). *Biometrika*, 52(3–4), 591–611. <https://doi.org/10.1093/biomet/52.3-4.591>
- Steiger, J. H., & Fouladi, R. T. (1997). Noncentrality interval estimation and the evaluation of statistical models. In L. L. Harlow, S. A. Mulaik, & J. H. Steiger (Eds.), *What if there were no significance tests?* (pp. 221–257). Lawrence Erlbaum.
- Tabachnick, B. G., & Fidell, L. S. (2019). *Using multivariate statistics* (7th ed.). Pearson.
- Welch, B. L. (1951). On the comparison of several mean values: An alternative approach. *Biometrika*, 38(3/4), 330–336. <https://doi.org/10.2307/2332579>
- Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6), 80–83. <https://doi.org/10.2307/3001968>